Check for updates

ACM DIGITAL LIBRARY | Association for Computing Machinery | acm open

Latest updates: https://dl.acm.org/doi/10.1145/3698061.3726924

RESEARCH-ARTICLE

Beyond Productivity: Rethinking the Impact of Creativity Support Tools

SAMUEL RHYS COX, Aalborg University, Aalborg, Nordjylland, Denmark



Samuel Rhys Cox is a researcher working in Human–Computer Interaction (HCI) and human-centred AI, with a focus on conversational systems and social computing. His research examines how interactive systems can be designed to help people feel more supported, particularly in sensitive or reflective contexts such as health, well-being, and creative practice. Through empirical studies and design-oriented research, he investigates how interaction design choices—such as conversational style, framing, and the use of prior interactions—influence users' experiences of comfort, reflection, and self-disclosure when interacting with AI systems.

HELENA BØJER DJERNÆS, Aalborg University, Aalborg, Nordjylland, Denmark

NIELS VAN BERKEL, Aalborg University, Aalborg, Nordjylland, Denmark

Open Access Support provided by:

Aalborg University

PDF Download
3698061.3726924.pdf
05 April 2026
Total Citations: 3
Total Downloads: 3074

Published: 22 June 2025

Citation in BibTeX format

C&C '25: Creativity and Cognition
June 23 - 25, 2025
Virtual, United Kingdom

Conference Sponsors:
SIGCHI

Beyond Productivity: Rethinking the Impact of Creativity Support Tools

Samuel Rhys Cox
Department of Computer Science
Aalborg University
Aalborg, Denmark
srcox@cs.aau.dk

Helena Bøjer Djernæs
Department of Computer Science
Aalborg University
Aalborg, Denmark
hbd@cs.aau.dk

Niels van Berkel
Department of Computer Science
Aalborg University
Aalborg, Denmark
nielsvanberkel@cs.aau.dk

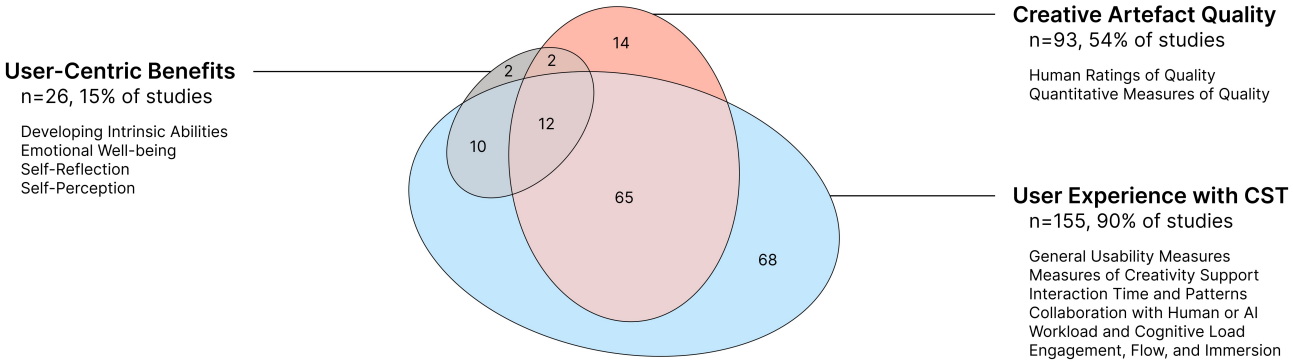


Figure 1: Measures used in (*n=173*) empirical studies of CSTs from a survey of 10 years of ACM DL literature. Measures of *User Experience with CSTs* were most prevalent (90%), followed by measures of *Creative Artefact Quality* (54%), and measures of *User-Centric Benefits* were least prevalent (15%).

Abstract

Creativity Support Tools (CSTs) are widely used across diverse creative domains, with generative AI recently increasing the abilities of CSTs. To better understand how the success of CSTs is determined in the literature, we conducted a review of outcome measures used in CST evaluations. Drawing from (*n=173*) CST evaluations in the ACM Digital Library, we identified the metrics commonly employed to assess user interactions with CSTs. Our findings reveal prevailing trends in current evaluation practices, while exposing underexplored measures that could broaden the scope of future research. Based on these results, we argue for a more holistic approach to evaluating CSTs, encouraging the HCI community to consider not only user experience and the quality of the generated output, but also user-centric aspects such as self-reflection and well-being as critical dimensions of assessment. We also highlight a need for validated measures specifically suited to the evaluation of generative AI in CSTs.



This work is licensed under a Creative Commons Attribution 4.0 International License.
C&C '25, Virtual, United Kingdom
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1289-0/25/06
https://doi.org/10.1145/3698061.3726924

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**.

Keywords

Creativity, Creativity Support Tools (CSTs), Literature Review, Evaluation Measures, Outcome Measures

ACM Reference Format:
Samuel Rhys Cox, Helena Bøjer Djernæs, and Niels van Berkel. 2025. Beyond Productivity: Rethinking the Impact of Creativity Support Tools. In *Creativity and Cognition (C&C '25), June 23–25, 2025, Virtual, United Kingdom.* ACM, New York, NY, USA, 15 pages. https://doi.org/10.1145/3698061.3726924

1 Introduction

Creativity support tools (CSTs) have been widely adopted to scaffold, guide, and enhance the creation of creative artefacts. CSTs have been applied across diverse domains, including creative writing and storytelling [177, 220], idea exploration [102, 128], and product design [98, 225], to name just a few examples. However, due to the mixed nature of CSTs, where their use is both artistic [203] and focused on generation of content [70], the evaluation of such tools can be broad and varied in aim. The rapid advancement of generative AI has further complicated this landscape, prompting reflection on the evolving role of the human user in CST interactions [18]. As these technologies continue to grow in capability and accessibility,

a pressing question emerges: should there be a philosophical shift in how the outcomes of CSTs are conceptualised and measured?

Motivated by this, we analyse 10 years of literature from the ACM Digital Library, surveying outcomes reported in empirical user studies of CSTs. From a qualitative analysis of 173 papers, we separate these outcomes into themes of: (1) measures of *user experience* while using CSTs, (2) measures of *creative artefact quality*, and (3) *user-centric* measures of benefits to the *individual themselves* beyond the generation of the creative artefact (such as learning or well-being). As a result, we find that the majority of HCI CST research has focused on measures of user experience and creative artefact quality, with a paucity of studies measuring user-centric benefits.

From this, we generate a call to researchers and practitioners to focus on outcome measures that help the *user themselves* through the creative process, as an area of focus in need of added attention. Additionally, we highlight need for validated user experience measures to account for growing affordances and limitations of using generative AI within CSTs. While we acknowledge the importance of improving the quality of creative output, in this work we wish to draw attention to outcomes that improve the quality of life of those using CSTs.

2 Related Work

Creativity is commonly defined as the generation of artefacts that are both novel (i.e., perceived as unique or original) and useful (i.e., perceived as valuable or feasible) within their given context [149]. Creativity involves cognitive processes such as divergent thinking (the generation of multiple varied ideas), and convergent thinking (refining and selecting the most appropriate idea).

Building on this understanding of creativity, researchers in HCI have long explored how computational systems can support and enhance creative processes. Creativity support tools (CST) have a long history in HCI literature [61, 62, 154], and have gained renewed prominence with the increasing accessibility of AI technologies [144]. Within this study, we define CSTs as systems or strategies that can be used to enhance and assist creative tasks, such as exploration, ideation, or creation. These include digital tools such as Fujinami et al.'s CST for painting with physical objects [64], Zhang et al.'s argumentative writing support tool [226], and Ford et al.'s music composition tool for professional composers [60]. CSTs also encompass non-digital support tools, as explored by Lundquist et al. [121] in their study of how physical tools (e.g., pen and paper), compare to digital and hybrid ideation setups, from which they argue for a combination of both.

Providing relevant creativity support is highly dependent on the intended end-user. Many studies focus on how to support adults during creative processes, such as writing [72, 160, 178], drawing [82, 110, 151]) or ideation [90, 145, 156], and have done so both targetting beginners seeking to get started in a new creative field, as well as practitioners looking to enhance their capability and productivity. For instance, Hu et al. [84] investigated creativity support through a collaborative environment where an adult user and a robot took turns creating art using tangram pieces. The study measured dimensions of creativity such as flexibility, originality, and elaboration to assess the extent to which the system supported creative output. After the task, participants completed a questionnaire and took part in interviews to provide further insights into perceived helpfulness and usability. In contrast, studies focusing on supporting children's creative processes often emphasise learning-oriented contexts [77, 201]. For example, Yan et al. [215], developed a digital collage tool (for collecting and inspecting new aspects of nature) that encourages children to explore, learn and create with nature.

Beyond the learning outcomes of CST studies discussed above, engaging in creative activities has been shown to offer significant benefits for the user themselves. Fancourt and Finn [58] offer an extensive scoping review that was published by the World Health Organization. Based on a coverage of over 3000 studies they found: "*a major role for the arts in the prevention of ill health, promotion of health, and management and treatment of illness across the lifespan*" [58], with the review discussing benefits from creative activities such as creative writing, dance, and painting. More specifically, creative activities have been found to benefit learning [66, 96], emotional well-being [39, 93], mental health [58], mindfulness [43], self-efficacy [27, 93], and self-esteem [27]. For example, brief (45 minute) free art-making facilitated by an art therapist improved both people's emotions and self-efficacy [93]. A creative writing program was found to enhance both the self-efficacy and self-esteem of adolescents [27]. Finally, the type of creative activity has also been found to affect the extent of positive benefits, with Curry and Kasser [43] finding that structured colouring activities (i.e., colouring in a mandala) led to greater reductions in anxiety compared to free colouring.

Drawing this together, this review explores the outcome measures used in the evaluation of CSTs in prior work. Importantly, we differentiate our work from that of Remy et al. [154], who surveyed the methodological approaches (e.g., surveys, interviews, and observations) used in CST evaluations up to 2019. In contrast, our review focuses on the explicit outcome measures employed, examining the specific scales, metrics, and their prevalence across the literature.

3 Methodology

Our approach was influenced by prior literature reviews within HCI [153, 227]. Specifically, we followed the four stages of: (1) Define: define the exclusion and inclusion criteria; (2) Search: develop search query and search relevant source; (3) Selection: filter resulting papers based on our selection and exclusion criteria; and (4) Analysis: perform qualitative analysis (open coding) on the final 173 total included papers

3.1 Definition

In this section, we define the inclusion and exclusion criteria applied during the selection process, to filter out relevant research papers from the initial 1241 included papers.

3.1.1 Inclusion criteria. This literature review includes studies that empirically evaluate user interactions with CSTs.

Beyond this, we include studies that integrate creative interaction and evaluate creativity-related outcomes, even when creativity is not the *primary* focus of the tool. For example, '*SelVReflect*' [185] employs a guided 3D drawing process in virtual reality to support creative self-expression as part of a reflective practice.

Additionally, to allow for a more broad surveying of creativity literature (and thereby a more holistic view of measures used), we included studies that use non-digital creativity tools, or that use forms of feedback to facilitate human-human creativity tasks. For example, Ma et al. developed an interface that visualises the gaze of another user during a drawing task on a shared canvas [122].

As the incorporation of AI into CSTs has increased, there have been competing definitions surrounding CSTs. While some prior work differentiates "*human-AI co-creativity*" from CSTs (due to AI being framed as a partner to creativity rather than a tool [156]), we follow the definition of prior work that considered human-AI co-creativity as a form of CST [18, 68], and the finding that CSTs could form part of a support network to creative output [38]. That is to say, co-creative AI can be seen to facilitate ideation and provides iterative feedback aligning with CST principles.

3.1.2 Exclusion criteria. Studies were excluded from the literature review, based on whether and how they evaluated a CST.

We excluded studies of tools that automatically generated content that was not subsequently edited by the user. However, we included studies if the CST automatically generated some forms of content that was then editable and allowed for user interactions (e.g., generation and editing of webpage layouts [22]).

Likewise, we excluded studies that developed taxonomies of user groups (i.e., categories for how users behave, rather than explorations of how people interact with a specific support tool). For example, Xu et al. generated four different patterns of behaviour used when people use search engines to help with creativity tasks [211], and Palani et al. generated roles that people follow when using generative AI [143].

We also excluded studies where creativity was primarily influenced through cognitive prompts rather than interactive or iterative scaffolding. For example, Yen et al. [218] examined how different reflection conditions affected iterative design revisions, but did not introduce or evaluate a creativity support tool (CST). While reflection can influence creativity, it does not provide structured, external scaffolding or interactive support, distinguishing it from CSTs that actively guide or modify creative outputs.

Studies that include surveys of how people interact with and perceive existing CSTs were also excluded (e.g., [28]), as they do not evaluate a specific tool within the study context. This includes online surveys reporting general usage trends of tools without presenting tool-specific evaluations.

We excluded papers that assessed interventions influencing creativity rather than evaluating a creativity support tool (CST) designed to support creativity directly. For example, Elgarf et al. [51] studied how regulatory focus affects children's verbal creativity using a social robot, but creativity was measured as an outcome of priming rather than as a function of the tool itself.

To summarise, we excluded papers based on the following criteria: (1) studies reporting only formative or needs-finding work, without evaluating a CST; (2) studies that excluded human involvement in the creative process (e.g., fully computer-generated outputs); (3) studies focused on developing taxonomies of user groups; (4) studies of automation with only human verification (e.g., [214]); (5) studies collecting only user sentiment (e.g., likes/dislikes [195]); (6) studies relying solely on observational data; (7) studies with unspecified or insufficient descriptions of outcome measures; (8) studies where creativity was shaped primarily by cognitive prompts, rather than interactive or iterative scaffolding; (9) studies presenting content-mining pipelines rather than interactive CSTs (e.g., [16, 41]); (10) studies focused on general interaction or perception of existing CSTs without creative outcomes; (11) studies evaluating interventions rather than CSTs. Unlike prior reviews that applied citation or download count thresholds [61, 62], we imposed no such restrictions in order to provide a broader overview of outcome measures.

Year	Publication
2024	[7, 10, 11, 17, 25, 29, 31–33, 36, 46, 57, 60, 65, 71, 79, 81, 81, 82, 92, 94, 97, 115, 127, 137, 140, 146, 152, 155, 158, 163, 172, 174–176, 184, 187, 189, 190, 197, 201, 204, 208, 213, 217, 219, 221, 223–225, 228, 229]
2023	[9, 23, 30, 37, 45, 59, 76, 80, 83, 85, 88, 91, 101, 110, 117, 133, 145, 147, 151, 171, 185, 188, 191, 193, 194, 199, 205, 210, 215, 216, 226]
2022	[2, 53, 55, 67, 102, 116, 156, 159, 177, 180, 183, 222]
2021	[35, 52, 63, 84, 86, 90, 95, 118, 119, 123, 136, 160, 178, 212]
2020	[4, 5, 13, 19, 22, 47, 106, 107, 114, 135, 141, 164, 165, 179, 186, 220, 231, 232]
2019	[1, 3, 12, 70, 105, 125, 128, 132, 134, 150, 167, 202, 203]
2018	[64, 121, 139, 192, 206]
2017	[8, 21, 24, 49, 54, 69, 72, 74, 98, 126, 138, 148, 166, 200, 207]
2016	[26, 42, 129, 169, 182]
2015	[48, 50, 73, 99, 100, 162, 168, 170]

Table 1: List of included papers (sorted by year).

3.2 Search

For this review, we surveyed research published in the ACM Digital Library in a 10 year time period from beginning of 2015 to end of 2024. When searching the ACM Digital Library, we followed prior reviews of CSTs [61, 62, 154] by searching for "*creativity support tool*" in any part of a publication. We also searched for papers that used "*creativity*" as a keyword to capture studies that evaluated CSTs, but did not necessarily include the phrase in their publication. This search resulted in a total of 1241 papers included in the initial selection process.

3.3 Selection and Analysis

To apply the inclusion and exclusion criteria, one researcher manually reviewed all 1241 papers, initially screening titles and abstracts, and consulting the introduction, methodology, and results sections where necessary. During this initial screening, any exclusion-relevant criteria not specified in the original list (e.g., papers that only collected user sentiment) were noted and discussed with the wider research team. The exclusion criteria were then revised to

reflect these additions and applied throughout the remainder of the screening process. This process resulted in 235 papers being included for outcome measure analysis.

Next, following prior work [153, 227] we conducted a thematic analysis of the remaining 235 papers. Two researchers qualitatively analysed the papers to identify and extract reported outcome measures. An open coding process was used to extract measures, with continuous discussion between coders to ensure consistency. Each paper was read in full to support detailed extraction of outcome measures. For example, Jamplate [213] was coded as using NASA-TLX and SUS (among other measures) to evaluate the publication's CST. In this process, we focused solely on outcome measures, and did not include any moderating or control variables (e.g., intrinsic motivation and creative thinking ability [150]).

During coding, an additional 62 papers were excluded, resulting in a final set of 173 papers included in the review. These were excluded due to insufficient reporting detail (e.g., vague descriptions of ad hoc measures with no cited source) or because they met exclusion criteria upon closer examination.

After completing qualitative analysis for all 173 papers, the two researchers consolidated codes into themes and subthemes. This process yielded three overarching themes: *User Experience with CST*, *Creative Artefact Quality*, and *User-Centric Benefits*. Codes were further organised into subthemes within each major theme. For instance, codes such as "*SUS*", "*UEQ*", "*usefulness (ad hoc)*" and "*satisfaction (ad hoc)*" measures were grouped under the "*General Usability Measures*" subtheme within the "*User Experience with CST*" theme.

Table 1 presents all 173 included papers sorted by publication year. Table 2 shows publications by venue, with CHI (*n=55*), C&C (*n=28*), DIS (*n=16*), and UIST (*n=16*) as the most represented.

Publication Venue	Paper Count
CHI: Conf. on Human Factors in Computing Systems	55
C&C: Creativity and Cognition	28
DIS: Designing Interactive Systems	16
UIST: User Interface Software and Technology	16
CSCW: Computer Supported Cooperative Work	8
CHI PLAY: Computer-Human Interaction in Play	5
ICMI: Int. Conf. on Multimodal Interaction	4
IUI: Intelligent User Interfaces	4
CI: Collective Intelligence	3
TEI: Tangible, Embedded and Embodied Interaction	3
TOCHI: Trans. on Computer-Human Interaction	3
Other	28

Table 2: Number of included papers (sorted by venue count). The category "Other" includes publication venues with two or less total publications.

For reader context, the reviewed literature spans a variety of application areas (though these were not part of our formal analysis). These include education, creative practice, industry, entertainment, and accessibility. In education, target users range from kindergarten to university students, encompassing a broad age spectrum. Industry-focused studies include live coders, designers, and writers, with focus on supporting ideation, design, collaboration, and content generation. Studies on creative practice target both casual users engaging in creative activities for leisure, as well as professional artists, including musicians, painters, and creative writers. In entertainment, target users include general audiences as well as professionals such as live-streamers and film-makers. Accessibility-focused work includes support for users with low vision, blindness, or aphasia.

4 Findings

We report the findings of our thematic analysis of the outcome variables contained in the *n=173* papers. Here, we structure our findings around the three high-level themes that emerged from our thematic analysis. Specifically, themes covered the following outcome measures:

- **User Experience with CST** (Section 4.1): Outcomes measuring the user experience during the ideation process when interacting with the CST, such as system usability.
- **Creative Artefact Quality** (Section 4.2): Outcomes measuring the quality of the creative output such as the quality of creative writing, music, or artist work.
- **User-Centric Benefits** (Section 4.3): Outcomes that benefit the *user themselves* outside of the CST, such as learning experiences, improvement in well-being and self-confidence.

Additionally, we report on the distribution of papers across subthemes within each main theme (e.g., "*Self-Reflection*" within the "*User-Centric Benefits*" theme). Table 3 provides counts of papers by theme and sub-theme. Figure 2 shows the number of themes per year. The rise in CST evaluations (see Fig. 2 and Tab. 1) is in part due to the increase in writing support tools and adoption of LLMs in HCI research [144].

Themes	Sub-themes	Paper Count
User Experience with CST		**155**
	General Usability Measures	101
	Measures of Creativity Support	58
	Interaction Time and Patterns	52
	Collaboration with Human or AI	36
	Workload and Cognitive Load	26
	Engagement, Flow, and Immersion	15
Creative Artefact Quality		**93**
	Human Ratings of Quality	74
	Quantitative Measures of Quality	54
User-Centric Benefits		**26**
	Developing Intrinsic Abilities	13
	Emotional Well-being	6
	Self-Reflection	6
	Self-Perception	5

Table 3: Outcome measures that emerged from the analysis. As studies use measures across multiple themes, sums will exceed total number of papers.

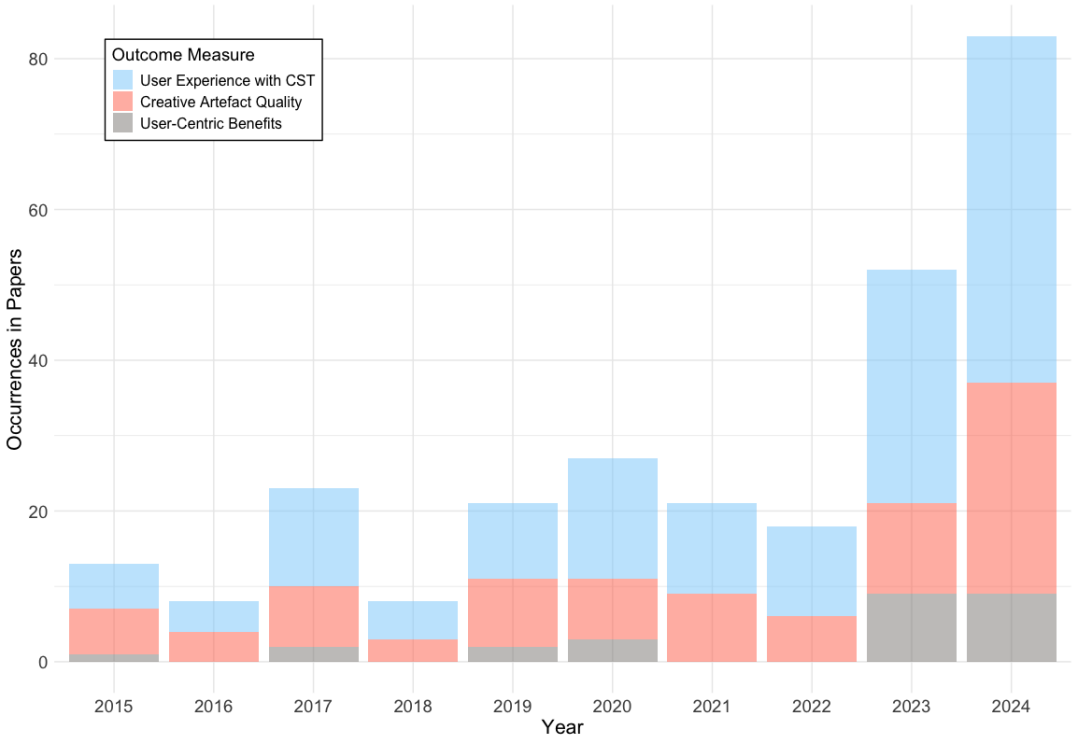


Figure 2: Count of measures used (User Experience with CST/Creative Artefact Quality/User-Centric Benefits) by year (2015–2024). As some studies use measures across multiple themes, sums will exceed total number of papers.

4.1 User Experience with CST

Of the 173 total papers, 155 evaluated the user experience of idea generation using the CST.

4.1.1 General Usability Measures: Here, the majority of studies (*n=101*) evaluated the general usability and user experience of the CST. Specifically, *n=84* studies incorporated ad hoc usability measures, *n=20* studies adopted the System Usability Scale (SUS) [20], *n=4* studies adopted the Usability Questionnaire (USE) [120], and *n=3* studies adopted the User Experience Questionnaire (UEQ) [109][1]. Ad hoc usability measures were typically used to evaluate specific question items (for example, ad hoc ease of use and enjoyment measures for the CST [135]) or to evaluate the usability of individual features within the CST. Additionally, some studies used combinations of validated measures (such as SUS) in combination with ad hoc usability measures, such as items to rate the ease of use of individual CST features (e.g., [47]).

4.1.2 Measures of Creativity Support: Measures specifically tailored to evaluating creativity support were used by *n=58* studies. Of these, *n=50* studies incorporated all or partial dimensions of the Creativity Support Index (CSI) [34][2], *n=5* studies used ad hoc measures, *n=2* studies used the Creativity Experience Self Rating Questionnaire (CESR) [44], *n=1* study used a scale (i.e., [130]) to measure how a virtual reality environment itself fosters creativity [94], and *n=2* studies used the Mixed-Initiative Creativity Support Index (MICSI) [110]. The MICSI adds a dimension related to human-AI co-creation to the CSI measuring the perceived control and communication with a co-creative AI partner (see Antony et al. for a creative writing CST evaluated using MICSI [10]).

4.1.3 Interaction Time and Patterns: The interaction activity, time, or behaviour patterns was recorded and evaluated by *n=52* studies. Interaction time was reported by *n=37* studies with focus either framing interaction time as being more beneficial as a lower value for more efficiency (e.g., [204]) or where longer interaction time is desirable as an indicator for increased user engagement and consideration during ideation (e.g., [185]). User interactions with the CST (e.g., button presses, conversational turns taken) were reported by *n=31* studies, with goal of high or low interaction count following a similar motivation (i.e., either framing less interactions as desireable due to increased efficiency [194], or more interactions framed as desireable due to increased user engagement [81]). Patterns of behaviour were also analysed by a number of studies [91], and the quantity and proportion of tool function use was reported as a measure of function popularity among users (e.g., [80]). Additionally, multiple studies reported the acceptance of CST suggestions (such as via semantic similarity between final artefact and CST inspirations, or as use indicated acceptance [84, 136, 174]).

[1]For completeness, other validated measures used (each in two or less studies) include QUIS, SEQ, PSSUQ, and AttrakDiff.
[2]Note: While the CSI [34] has an Enjoyment subscale (with items such as: "*I would be happy to use this system or tool on a regular basis.*"), this is enjoyment of user experience and is therefore not coded as a user-centric outcome.

4.1.4 Collaboration with Human or AI:. Next, *n=36* studies used measures focusing on aspects of collaboration. These included measures of human-human collaboration (where a CST facilitated collaborative creativity), user perceptions of CST-generated feedback, and interactions specifically involving AI components within CSTs. Here, there was some crossover, with the same outcome measures being used to evaluate both human-human collaboration and human-AI co-creativity. For example, earlier crowdsourcing or group collaboration studies often used measures of group cohesion or levels of agreement between group members (e.g., [1, 132]). On from this, recent work has used similarly motivated question items to evaluate interactions with human-AI co-creativity CSTs (e.g., [86, 133, 171]. However, while human-human studies may use validated scales, human-AI partnerships more often used ad hoc measures (e.g., "*I felt like I was collaborating with the AI system*" [133]) indicating a potential need for more research into validated measures for human-AI co-creativity.

This is particularly pressing given the rising capabilities and limitations surrounding the use of LLMs in creative work. Here, CSTs can introduce issues surrounding hallucinations, lack of consistency and predictability, as well as perceptions surrounding perceived ownership of created artefacts and perceived agency and control when working in a human-AI partnership. Demonstrating this breadth of potential issues and need for added measures are recent works that adopt ad hoc measures to account for emerging concerns with human-LLM interactions (e.g., [133, 152, 155, 197] where no validated measure for perceived ownership or perceived agency emerged).

4.1.5 Workload and Cognitive Load: Workload and cognitive load were evaluated by *n=26* studies, with *n=19* studies using the NASA Task Load Index [78], *n=5* studies using ad hoc measures, and *n=2* studies using other validated scales. Here, ad hoc measures were used to generate single item measures specific to the use case of the CST. For example, Petridis et al. recruited journalists to evaluate AngleKindling, a CST to support journalistic ideation, and used the ad hoc item ("*Coming up with story ideas was mentally taxing with this system.*") to measure mental demand [147].

4.1.6 Engagement, Flow, and Immersion Measures: User engagement, flow (i.e., the feeling of losing time), and immersion was evaluated by *n=15* studies. For example, Schlagowski et al. used the Flow Short Scale to measure participants' flow while using a music production CST [158]. Additionally, some studies used relative subjective duration (RSD) flow measure to quantify a difference between a participant's perceived interaction time and the actual interaction time (e.g., [182, 232]). Measures of flow had crossover with scales used for video games, such as the Game Experience questionnaire, where some studies used all dimensions or flow dimension only (e.g., [220, 232]). Here, only *n=1* study used ad hoc measures, demonstrating the current fulfilment of validated measures.

4.2 Creative Artefact Quality

Of the 173 total papers, 93 evaluated the quality of the artefact produced using the CST. For example, studies evaluated the quality of innovative product designs [223], the perceived humour of slogans [97], or the expressiveness of artwork [82]. Evaluating artefact quality generally relied on human ratings, or measures of quality derived via quantitative content analysis. Within this theme, well established measures of creativity were generally adopted, with measures such as *fluency*, *flexibility*, *originality* and *elaboration* receiving wide use.

4.2.1 Human Ratings of Artefact Quality: For human ratings, one popular approach (*n=34*) was for ideators to rate the perceived quality or satisfaction of the artefact they themselves had created[3]. For example in Metaphorian, ideators were asked to rate the accuracy, originality and coherence of their scientific metaphors [101], and Zhong et al. asked ideators to rate the perceived usefulness and correctness of diagrams [228]. Within the *Creative Artefact Quality* theme, these self-rated measures were often the most ad hoc[4] with authors creating custom measures to match the particular use case of their study.

Human ratings were also provided by third parties such as experts and laypeople (*n=48*). While ad hoc measures were still used, more of these ratings were structured around existing theory or techniques. For example, Amabile's Consensual Creativity Assessment Technique [6] was adopted by several studies to guide ratings (e.g., [32, 202, 222]). Another method was to derive measures from the Torrance Thinking Creativity Test (TTCT), such as evaluations for creative writing [72] or drawings [82]. As a common definition of creativity is for an artefact to be both novel and feasible, these were commonly used as assessment criteria for raters who were both experts and laypeople [1, 8, 26, 94, 169].

4.2.2 Quantitative Measures of Artefact Quality: Additionally, artefacts were evaluated using quantitative approaches (*n=54*). Commonly CSTs were evaluated using the four indicators of divergent thinking from TTCT (i.e., fluency, flexibility, elaboration, and originality).

Here, fluency (the number of ideas) was adopted by a sizeable number of studies (*n=34*). The precise definition of fluency within publications was similar, but with varying levels of precision. While some work reported fluency as the total number of ideas generated during the study, other work followed a more rigorous approach such as by excluding incomplete or irrelevant ideas [26], or by including only unique ideas [17]. Further, Zhang et al. used verbal analysis methods to analysis the fluency of children's utterances during visual storytelling tasks [222].

Flexibility, the number of unique categories across all ideations, was used by *n=13* studies (studies also defined this outcome as *breadth* or *variety* while applying the same high-level definition). Flexibility was often calculated by manually coding all ideas to generate categories [7, 70], or by quantifying mind maps [13, 69]. On from this, *n=6* studies analysed the number of ideas per category referring to either the *depth* or *persistence* of ideas [13, 17, 26, 69, 94, 150].

Elaboration, the amount of detail in responses, was reported by *n=15* studies. Here, elaboration was often reported using word

[3]As a caveat, while some studies framed self-rated satisfaction in outcome as a measure of quality of artefact [13, 95, 180, 210], in other work similarly worded measures were included as part of general system usability measures. Here we took an approach to include measures as self-ratings of quality if the item explicitly referenced the artefact that was generated (e.g., "[...] *satisfied with **your painting***" [210]).
[4]By "*ad hoc*" measures, we refer to measures that were generated by the researchers themselves for the purpose of their study.

count [7]. Additionally, some work analysed the content of ideas (for example the "*when, where and why*" level of detail in stories written by children [221], or the level of expressivity in children's verbal storytelling utterances [52]).

Originality, the rarity of a response, was reported by *n=15* studies. Here, studies calculated originality computationally (e.g., [13, 48]), by manually coding responses (e.g., [84]) or by via human ratings (e.g., [53, 86]).

Finally, *n=10* studies evaluated the diversity of ideas produced. Diversity as a concept has various definitions and methods of calculation [41], but can typically be conceptualised as the coverage of ideas *collectively* over an idea space [169]. For written artefacts, sentence embeddings were used to calculate the semantic similarity between ideas (e.g., [1, 7]). Here, the general motivation was to increase the diversity of ideas generated during idea exploration. While there is some overlap between diversity and the depth and breadth of ideas, diversity related studies followed a less categorical approach to quantifying ideas.

4.3 User-Centric Benefits

Of the 173 total papers, 26 reported measures that benefit the *user themselves* as a result of using the CST. By this, we refer to measures that are aimed towards empowering the user's intrinsic abilities, feelings, self-confidence and well-being rather than the quality of creative artefact or user experience[5]. User-centric measures often deployed both pre- and post-tests to measure change. Additionally, user-centric measures were particularly prevalent within studies where children were the user group (*n=10*), especially when learning outcomes were assessed.

4.3.1 Developing Intrinsic Abilities: Measures that focused on developing people's intrinsic abilities were used in *n=13* studies. Here, studies assessed outcomes such as learning of concepts and terminology, as well as improvements in creative skills, often using pre- and post-tests to evaluate change. Learning outcomes were measured by *n=11* studies. For example, improving AI literacy among children [201] and designers [146], helping designers learn to identify issues in visual design [219], or learning of domain specific terminology through creative tasks [88, 221].

Additionally, two studies [4, 5] used the "Test for Creative Thinking-Drawing Production" (TCT-DP) [89] to evaluate the creativity of children's drawings after using a CST. In this test, children are asked to complete an incomplete drawing within an (unknown to the children) time limit. The resultant drawings are then scored across multiple dimensions to assess creativity. Alves-Oliveira et al. assessed children's verbal creativity using pre- and post-test divergent thinking tasks [5]. In these tasks, children viewed an image and verbally asked as many questions as they could, with responses being analysed to produce a single verbal creativity score. Finally, Engelman et al. [54] developed survey items based on prior creativity literature to assess traits related to creative expressiveness, exploration, immersion, and thinking skills. These items were used in pre- and post-tests to evaluate potential changes in students' creativity.

[5]Note: We differentiate user-centric outcome measures from moderating or control variables. For example, intrinsic motivation and creative thinking ability was used by Przybilla et al. as a control variable and therefore not included in this theme [150].

4.3.2 Emotional Well-being: People's emotional well-being as a result of CST interaction was measured by *n=6* studies[6]. Several studies used adjective-based scales, featuring items such as "*proud*", "*upset*", or "*happy*". Participants either rated the extent to which they experienced each emotion using validated scales (such as PANAS [196] or AD-ACL [181]), or selected the adjectives that best described their mood (based on ad hoc measures [73, 74]), both before and after the interaction.

For example, Gonçalves et al. [73] evaluated creative writing support tools with marginalised youth, measuring well-being after each daily writing session over a two-week period to track changes in well-being throughout the study. Wagener et al. [185] used the PANAS scale to assess participants' emotions before and after interacting with their CST, in order to evaluate its effect on positive and negative affect. Finally, Wan et al. [187, Table 1] designed questions using Mekler and Hornbæk's framework for the experience of meaning [131], many of which are closely tied to emotional well-being through their focus on users' emotional responses, sense of connection, and personal resonance with the system.

4.3.3 Self-Reflection: Finally, *n=6* studies evaluated measures of self-reflection[7], with studies using both ad hoc and validated measures. One validated measure used was the Reflection in Creative Experience (RiCE) questionnaire [59, 60], a 9-item scale to measure reflection occurring in creative practice. RiCE includes a subscale specifically focused on reflection on self, with items such as: "*I learned many new things about myself during the experience*". Another validated measure used was the Technology-Supported Refection Inventory (TSRI) [15], a 9-item scale to measure how well a system supports reflection. TSRI features subscales for insight, exploration, and comparison, with items such as: "*Using the system gives me ideas on how to overcome challenges*".

For instance, Wagener et al. developed '*SelVReflect*' [185] to guide participants to self-reflect during an assisted drawing task, and measured self-reflection using an adjusted form of the TSRI. Yan et al. designed '*NaCanva*' [215], a CST to help children develop mood boards to aid in personal reflection (i.e., recalling memories). '*NaCanva*' also aimed to promote learning about nature, and foster a positive attitude and engagement with nature (such as improving acting and caring on nature, awareness and emotional ties to nature, and feelings of identity within nature).

4.3.4 Self-Perception: Measures related to the self-perception of participants were included in *n=5* studies. Specifically, studies measured participants' self-efficacy[8] (e.g., [30, 54, 145, 185]), and sense of achievement (e.g., [30, 210]). All studies measuring self-efficacy used both pre- and post-tests, with scales used including the General Self-Efficacy Scale (GSE) [161] (e.g., [185]), and measures derived from existing scales (e.g., [54]).

[6]Note: While studies have examined regulating ideators' emotions to support creativity [48, 50], these emotions are primarily tied to the user experience of the CST itself (e.g., frustration or satisfaction during use [48]). Therefore, we do not include these studies under the theme of user-centric benefits, and instead focus on emotional outcomes that more explicitly benefit the user beyond their interaction with the CST.
[7]Note: The term "*self-reflection*" refers specifically to the user's *personal* reflection. As such, reflection on the CST interaction itself is not included.
[8]"*Self-efficacy*" includes conceptually similar measures such as "*creative self-belief*", and "*self-confidence*".

For example, Park et al. developed the CST '*Exergy*' to support non-experts in generating sustainable energy ideas [145]. Perceived difficulty and technological confidence were measured pre- and post-interaction, aiming to improve users' confidence and make sustainable solutions more accessible to non-experts. Furthermore, Chavula et al. measured feelings of accomplishment after idea generation using CSTs [30], and Xu et al. measured participants' sense of achievement after creating a painting [210].

5 Discussion

In this paper, we conducted a review of outcome measures used to evaluate user studies of CSTs. From our review, we highlight three emerging themes of outcomes, and draw attention to predominant outcomes in prior work, and highlight areas that warrant further attention from the HCI and creativity communities. Specifically, we found that *User Experience with CSTs* in roughly 90% of studies, *Creative Artefact Quality* was measured in roughly 54% of studies, and *User-Centric Benefits* in only 15% of studies.

First, we would like to contextualise this review against the rapidly growing capabilities of human-AI creativity partnerships [157], whereby LLMs (and other generative-AI methods) can simulate and replicate detailed human-like outputs [18]. On from this, as creativity as a process adopts the use of AI-generated artefacts, the need for human creativity may be seen as unnecessary if one's perspective is framed around the quality of a creative artefact itself. In other words, if generative-AI can produce human-like output, then will the need for human input to creative pursuits diminish in return? Motivated by this, we investigated the use outcome measures in CSTs historically within HCI literature, with aim to gain insight into the popularity and validity of prior used metrics.

Evaluations of user experience (UX) commonly incorporated general measures (such as understandability or ease of use), with the use of ad hoc measures being predominant. In part, ad hoc measures were developed to evaluate UX within novel CSTs and their accompanying features. However, such use of ad hoc measures also draws attention to the need for increased focus on creating validated measures tailored to CST evaluation. The use of ad hoc measures also reduces the ability to reproduce work (with some studies being insufficiently described in terms of outcome measures and therefore not suitable for inclusion in our review).

Additionally, recent CST evaluations have included the use of ad hoc measures that account for the added affordances and limitations of generative AI [86, 133, 171]. Alongside the rapid emergence of generative-AI for real-world creativity tasks, the use of validated measures that account for these affordances and potential limitations is needed within the creativity and HCI community. Additionally, we would like to highlight the lack of measures adopted in studies that would be present within the conversational user interface (CUI) community (see reviews for example measures used within CUI literature [153, 227]). For example, measures have been used to evaluate the quality of conversations between users and CUIs [40, 75, 87, 209], to evaluate user-centric benefits of conversations [108, 230], and validated scales of conversational agents have been used [173, 198] (such as Godspeed [14, 112]). As conversational interactions (such as with co-creative AI) becomes more prevalent, need for such validated measures will increase.

Measures of creative artefact quality were fairly consistent and established within the literature, with measures derived from creativity literature allowing for rigour, consistency, and reproducibility. These measures included both human ratings of creative artefacts and quantitative content analyses. Additionally, measures were commonly derived from the Torrance Thinking Creativity Test (TTCT). The prevalence of such TTCT related measures reflects the popularity of divergent thinking tasks within CSTs, where historically the exploration of ideas has seen increased attention compared to convergent thinking and idea refinement [63].

Finally, our review highlights a notable lack of measures that evaluate benefits to the *users themselves*, such as assessments of intrinsic abilities, emotions, self-confidence, and overall well-being. Although recent work has highlighted a motivational shift towards user-centric outcomes (such as empowerment within artistic support tools [111], "*slowing down*" for well-being and relaxation [56], or playful engagement within CSTs [113]) there remains a need for further research explicitly addressing these user-centric dimensions. This need is further supported by insights from creative practitioners, who have emphasised emotional connection (e.g., feelings of happiness and belonging) as a valued aspect of CSTs [142]. The limited use of user-centric measures may come as additionally surprising given the well-established psychological and emotional benefits of creative activity [43, 58, 93]. The creativity literature (particularly the extensive scoping review by Fancourt and Finn [58]) offers a strong foundation to inform the design and evaluation of future CSTs in HCI. Moreover, as behaviour change is inherently a long-term process, evaluating outcomes such as well-being or stress reduction requires longitudinal study designs in order to reliably assess the effectiveness of CSTs [103, 104]. In light of these motivations, we encourage greater attention to user-centric outcomes, in addition to the current emphasis on user experience and artefact quality in CST evaluation.

5.1 Limitations

Our survey covered 10 years of literature from the ACM Digital Library. While the ACM Digital Library has been used in prior reviews as a representative sample of computing literature [61, 62, 154, 227], the inclusion of additional academic repositories (such as IEEE Xplore or Science Direct) may uncover insights or measures not contained in our review, and overcome potential selection bias from the sole use of ACM publications. Nevertheless, given that our survey systematically analysed a sample of *n=173* papers spanning a 10-year period (comparable to or exceeding the scope of many prior reviews) we argue that our findings provide a robust and representative foundation for further inquiry.

We would like to highlight that the definition of a Creativity Support Tool (CST) is not always consistent. On from this, we took the decision to include co-creative AI (such as LLM support [155] and chatbot partners [86]). By taking this broader approach, our review captures a wider range of tools that support creativity, ensuring a more representative analysis of the field. Additionally, we did not include exploratory studies (e.g., [124]) in our review of outcome measures (as such studies do not include explicitly defined outcome measures by design). While we acknowledge that measures of more user-centric benefits may be present in more exploratory studies,

we provoke that the paucity of such measures as predefined outcomes highlights a growing need of focus for the HCI and creativity community.

6 Conclusion

This paper reviewed the outcome measures used in the evaluation of CSTs as published in the ACM Digital Library, via an analysis of 173 papers as published over a ten year period (2015–2024). Our results show that the user experience of using the CST was assessed in close to 90% evaluated studies. Just over half of the assessed studies explicitly evaluated the quality of the creative output. Finally, user-centric benefits were evaluated by only 15% of the studies included in our review.

CSTs are increasingly effective in generating both a large quantity and high quality of output, largely enabled through the use of generative AI. This inevitably results in changes to the creative process and benefits to the users of CSTs. We therefore urge the community to consider a more holistic approach to evaluating CSTs and carefully consider the impact of these tools on user-centric benefits. Additionally, we highlight the need for validated measures specifically suited to the evaluation of generative AI in CSTs to overcome the current reliance on ad hoc measures.

Acknowledgments

We would like to thank our reviewers for their positive reception of the work, and their helpful and construction feedback to improve the paper. This work is supported by the Carlsberg Foundation, grant CF21-0159.

References

[1] Faez Ahmed, Nischal Reddy Chandra, Mark Fuge, and Steven Dow. 2019. Structuring Online Dyads: Explanations Improve Creativity, Chats Lead to Convergence. In *Proceedings of the 2019 Conference on Creativity and Cognition*. 306–318. doi:10.1145/3325480.3325486

[2] Safinah Ali, Nisha Elizabeth Devasia, and Cynthia Breazeal. 2022. Escape!Bot: Social Robots as Creative Problem-Solving Partners. In *Proceedings of the 14th Conference on Creativity and Cognition*. 275–283. doi:10.1145/3527927.3532793

[3] Safinah Ali, Tyler Moroso, and Cynthia Breazeal. 2019. Can Children Learn Creativity from a Social Robot?. In *Proceedings of the 2019 Conference on Creativity and Cognition*. 359–368. doi:10.1145/3325480.3325499

[4] Safinah Ali, Hae Won Park, and Cynthia Breazeal. 2020. Can Children Emulate a Robotic Non-Player Character's Figural Creativity?. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*. 499–509. doi:10.1145/3410404.3414251

[5] Patrícia Alves-Oliveira, Patrícia Arriaga, Matthew A Cronin, and Ana Paiva. 2020. Creativity Encounters Between Children and Robots. In *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*. 379–388. doi:10.1145/3319502.3374817

[6] Teresa M Amabile. 1982. Social psychology of creativity: A consensual assessment technique. *Journal of Personality and Social Psychology* 43, 5 (1982), 997. https://psycnet.apa.org/doi/10.1037/0022-3514.43.5.997

[7] Barrett R Anderson, Jash Hemant Shah, and Max Kreminski. 2024. Homogenization Effects of Large Language Models on Human Creative Ideation. In *Proceedings of the 16th conference on creativity & cognition*. 413–425. doi:10.1145/3635636.3656204

[8] Salvatore Andolina, Hendrik Schneider, Joel Chan, Khalil Klouche, Giulio Jacucci, and Steven Dow. 2017. Crowdboard: Augmenting In-Person Idea Generation with Real-Time Crowds. In *Proceedings of the 2017 ACM SIGCHI Conference on Creativity and Cognition*. 106–118. doi:10.1145/3059454.3059477

[9] Tyler Angert, Miroslav Suzara, Jenny Han, Christopher Pondoc, and Hariharan Subramonyam. 2023. Spellburst: A Node-based Interface for Exploratory Creative Coding with Natural Language Prompts. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–22. doi:10.1145/3586183.3606719

[10] Victor Nikhil Antony and Chien-Ming Huang. 2024. ID.8: Co-Creating Visual Stories with Generative AI. *ACM Transactions on Interactive Intelligent Systems* 14, 3 (2024), 1–29. doi:10.1145/3672277

[11] Ashratuz Zavin Asha, Mark Colley, Sharifa Sultana, and Ehud Sharlin. 2024. Introducing AV-Sketch: An Immersive Participatory Design Tool for Automated Vehicle—Passenger Interaction. In *Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. 83–93. doi:10.1145/3640792.3675705

[12] Ilhan Aslan, Katharina Weitz, Ruben Schlagowski, Simon Flutura, Susana Garcia Valesco, Marius Pfeil, and Elisabeth André. 2019. Creativity Support and Multimodal Pen-based Interaction. In *2019 International Conference on Multimodal Interaction*. 135–144. doi:10.1145/3340555.3353738

[13] Suyun Sandra Bae, Oh-Hyun Kwon, Senthil Chandrasegaran, and Kwan-Liu Ma. 2020. Spinneret: Aiding Creative Ideation through Non-Obvious Concept Associations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13. doi:10.1145/3313831.3376746

[14] Christoph Bartneck, Dana Kulić, Elizabeth Croft, and Susana Zoghbi. 2009. Measurement Instruments for the Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety of Robots. *International Journal of Social Robotics* 1 (2009), 71–81. doi:10.1007/s12369-008-0001-3

[15] Marit Bentvelzen, Jasmin Niess, Mikołaj P Woźniak, and Paweł W Woźniak. 2021. The Development and Validation of the Technology-Supported Reflection Inventory. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–8. doi:10.1145/3411764.3445673

[16] Bhavya Bhavya, Jinjun Xiong, and Chengxiang Zhai. 2023. CAM: A Large Language Model-based Creative Analogy Mining Framework. In *Proceedings of the ACM Web Conference 2023*. 3903–3914. doi:10.1145/3543507.3587431

[17] Nanyi Bi, Yueh-Chi Chi, Yun-Yun Lin, Min-Jui Lee, and Bing-Yu Chen. 2024. Blow Your Mind: Exploring the Effects of Scene-Switching and Visualization of Time Constraints on Brainstorming in Virtual Reality. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW2 (2024), 1–23. doi:10.1145/3687026

[18] Oloff C Biermann, Ning F Ma, and Dongwook Yoon. 2022. From Tool to Companion: Storywriters Want AI Writers to Respect Their Personal Values and Writing Strategies. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference*. 1209–1227. doi:10.1145/3532106.3533506

[19] Simon Bourdeau, Annemarie Lesage, Béatrice Couturier Caron, and Pierre-Majorique Léger. 2020. When Design Novices and LEGO® Meet: Stimulating Creative Thinking for Interface Design. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14. doi:10.1145/3313831.3376495

[20] John Brooke et al. 1996. SUS - A quick and dirty usability scale. *Usability Evaluation in Industry* 189, 194 (1996), 4–7.

[21] Andrew R Brown, Toby Gifford, and Bradley Voltz. 2017. Stimulating Creative Partnerships in Human-Agent Musical Interaction. *Computers in Entertainment (CIE)* 14, 2 (2017), 1–17. doi:10.1145/2991146

[22] Daniel Buschek, Charlotte Anlauff, and Florian Lachner. 2020. Paper2Wire: a case study of user-centred development of machine learning tools for UX designers. In *Proceedings of Mensch und Computer 2020*. 33–41. doi:10.1145/3404983.3405506

[23] Alice Cai, Steven R Rick, Jennifer L Heyman, Yanxia Zhang, Alexandre Filipowicz, Matthew Hong, Matt Klenk, and Thomas Malone. 2023. DesignAID: Using Generative AI and Semantic Diversity for Design Inspiration. In *Proceedings of The ACM Collective Intelligence Conference*. 1–11. doi:10.1145/3582269.3615596

[24] Anil Çamcı, Kristine Lee, Cody J Roberts, and Angus G Forbes. 2017. INVISO: A Cross-platform User Interface for Creating Virtual Sonic Environments. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology*. 507–518. doi:10.1145/3126594.3126644

[25] Tuhin Chakrabarty, Vishakh Padmakumar, Faeze Brahman, and Smaranda Muresan. 2024. Creativity Support in the Age of Large Language Models: An Empirical Study Involving Professional Writers. In *Proceedings of the 16th Conference on Creativity & Cognition*. 132–155. doi:10.1145/3635636.3656201

[26] Joel Chan, Steven Dang, and Steven P Dow. 2016. Improving Crowd Innovation with Expert Facilitation. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing*. 1223–1235. doi:10.1145/2818048.2820023

[27] Genevieve E Chandler. 1999. A creative writing program to enhance self-esteem and self-efficacy in adolescents. *Journal of Child and Adolescent Psychiatric Nursing* 12, 2 (1999), 70–78. doi:10.1111/j.1744-6171.1999.tb00047.x

[28] Minsuk Chang, Stefania Druga, Alexander J Fiannaca, Pedro Vergani, Chinmay Kulkarni, Carrie J Cai, and Michael Terry. 2023. The Prompt Artists. In *Proceedings of the 15th Conference on Creativity and Cognition*. 75–87. doi:10.1145/3591196.3593515

[29] Ruei-Che Chang, Yuxuan Liu, Lotus Zhang, and Anhong Guo. 2024. EditScribe: Non-Visual Image Editing with Natural Language Verification Loops. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–19. doi:10.1145/3663548.3675599

[30] Catherine Chavula, Yujin Choi, and Soo Young Rieh. 2023. SearchIdea: An Idea Generation Tool to Support Creativity in Academic Search. In *Proceedings of the 2023 Conference on Human Information Interaction and Retrieval*. 161–171. doi:10.1145/3576840.3578294

[31] Liuqing Chen, Qianzhi Jing, Yixin Tsang, Qianyi Wang, Ruocong Liu, Duowei Xia, Yunzhan Zhou, and Lingyun Sun. 2024. AutoSpark: Supporting Automobile Appearance Design Ideation with Kansei Engineering and Generative AI. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–19. doi:10.1145/3654777.3676337

[32] Liuqing Chen, Shuhong Xiao, Yunnong Chen, Yaxuan Song, Ruoyu Wu, and Lingyun Sun. 2024. ChatScratch: An AI-Augmented System Toward Autonomous Visual Programming Learning for Children Aged 6-12. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–19. doi:10.1145/3613904.3642229

[33] Qing Chen, Wei Shuai, Jiyao Zhang, Zhida Sun, and Nan Cao. 2024. Beyond Numbers: Creating Analogies to Enhance Data Comprehension and Communication with Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–14. doi:10.1145/3613904.3642480

[34] Erin Cherry and Celine Latulipe. 2014. Quantifying the Creativity Support of Digital Tools through the Creativity Support Index. *ACM Transactions on Computer-Human Interaction (TOCHI)* 21, 4 (2014), 1–25. doi:10.1145/2617588

[35] Yen-Ting Cho, Yen-Ling Kuo, Yen-Ting Yeh, Yen-Yi Huang, and Po-Lun Huang. 2021. IntuModels: Enabling Interactive Modeling for the Novice through Idea Generation and Selection. In *Proceedings of the 13th Conference on Creativity and Cognition*. 1–10. doi:10.1145/3450741.3465241

[36] DaEun Choi, Sumin Hong, Jeongeon Park, John Joon Young Chung, and Juho Kim. 2024. CreativeConnect: Supporting Reference Recombination for Graphic Design Ideation with Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–25. doi:10.1145/3613904.3642794

[37] John Joon Young Chung and Eytan Adar. 2023. Artinter: AI-powered Boundary Objects for Commissioning Visual Arts. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 1997–2018. doi:10.1145/3563657.3595961

[38] John Joon Young Chung, Shiqing He, and Eytan Adar. 2022. Artist Support Networks: Implications for Future Creativity Support Tools. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference*. 232–246. doi:10.1145/3532106.3533505

[39] Tamlin S. Conner, Colin G. DeYoung, and Paul J. Silvia and. 2018. Everyday creative activity as a path to flourishing. *The Journal of Positive Psychology* 13, 2 (2018), 181–189. doi:10.1080/17439760.2016.1257049

[40] Samuel Rhys Cox and Wei Tsang Ooi. 2022. Does Chatbot Language Formality Affect Users' Self-Disclosure?. In *Proceedings of the 4th Conference on Conversational User Interfaces*. 1–13. doi:10.1145/3543829.35438

[41] Samuel Rhys Cox, Yunlong Wang, Ashraf Abdul, Christian Von Der Weth, and Brian Y. Lim. 2021. Directed Diversity: Leveraging Language Embedding Distances for Collective Creativity in Crowd Ideation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–35. doi:10.1145/3411764.3445782

[42] Rui Craveirinha and Licinio Roque. 2016. Exploring the Design-Space: The Authorial Game Evolution Tool Case-Study. In *Proceedings of the 13th International Conference on Advances in Computer Entertainment Technology*. 1–10. doi:10.1145/3001773.3001778

[43] Nancy A. Curry and Tim Kasser and. 2005. Can Coloring Mandalas Reduce Anxiety? *Art Therapy* 22, 2 (2005), 81–85. doi:10.1080/07421656.2005.10129441

[44] Darren W Dahl and C Page Moreau. 2007. Thinking inside the Box: Why Consumers Enjoy Constrained Creative Experiences. *Journal of Marketing Research* 44, 3 (2007), 357–369. doi:10.1509/jmkr.44.3.

[45] Hai Dang, Frederik Brudy, George Fitzmaurice, and Fraser Anderson. 2023. WorldSmith: Iterative and Expressive Prompting for World Building with a Generative AI. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–17. doi:10.1145/3586183.3606772

[46] Richard Lee Davis, Thiemo Wambsganss, Wei Jiang, Kevin Gonyop Kim, Tanja Käser, and Pierre Dillenbourg. 2024. Fashioning Creative Expertise with Generative AI: Graphical Interfaces for Design Space Exploration Better Support Ideation Than Text Prompts. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–26. doi:10.1145/3613904.3642908

[47] Niraj Ramesh Dayama, Kashyap Todi, Taru Saarelainen, and Antti Oulasvirta. 2020. GRIDS: Interactive Layout Design with Integer Programming. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13. doi:10.1145/3313831.3376553

[48] Alwin de Rooij, Philip J Corr, and Sara Jones. 2015. Emotion and Creativity: Hacking into Cognitive Appraisal Processes to Augment Creative Ideation. In *Proceedings of the 2015 ACM SIGCHI Conference on Creativity and Cognition*. 265–274. doi:10.1145/2757226.2757227

[49] Alwin de Rooij, Philip J Corr, and Sara Jones. 2017. Creativity and Emotion: Enhancing Creative Thinking by the Manipulation of Computational Feedback to Determine Emotional Intensity. In *Proceedings of the 2017 ACM SIGCHI Conference on Creativity and Cognition*. 148–157. doi:10.1145/3059454.3059469

[50] Alwin de Rooij and Sara Jones. 2015. (E)Motion and Creativity: Hacking the Function of Motor Expressions in Emotion Regulation to Augment Creativity. In *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction*. 145–152. doi:10.1145/2677199.2680552

[51] Maha Elgarf, Natalia Calvo-Barajas, Patricia Alves-Oliveira, Giulia Perugia, Ginevra Castellano, Chirstopher Peters, and Ana Paiva. 2022. "And then what happens?" Promoting Children's Verbal Creativity Using a Robot. In *2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 71–79. doi:10.1109/HRI53351.2022.9889408

[52] Maha Elgarf, Gabriel Skantze, and Christopher Peters. 2021. Once Upon a Story: Can a Creative Storyteller Robot Stimulate Creativity in Children?. In *Proceedings of the 21st ACM International Conference on Intelligent Virtual Agents*. 60–67. doi:10.1145/3472306.3478359

[53] Maha Elgarf, Sahba Zojaji, Gabriel Skantze, and Christopher Peters. 2022. CreativeBot: a Creative Storyteller robot to stimulate creativity in children. In *Proceedings of the 2022 International Conference on Multimodal Interaction*. 540–548. doi:10.1145/3536221.3556578

[54] Shelly Engelman, Brian Magerko, Tom McKlin, Morgan Miller, Doug Edwards, and Jason Freeman. 2017. Creativity in Authentic STEAM Education with EarSketch. In *Proceedings of the 2017 ACM SIGCSE Technical Symposium on Computer Science Education*. 183–188. doi:10.1145/3017680.3017763

[55] Inan Evin, Perttu Hämäläinen, and Christian Guckelsberger. 2022. Cine-AI: Generating Video Game Cutscenes in the Style of Human Directors. *Proceedings of the ACM on Human-Computer Interaction* 6, CHI PLAY (2022), 1–23. doi:10.1145/3549486

[56] Jeanette Falk, Daniel Spikol, Diana Meda, and Mark Melnykowycz. 2024. The Role of Slowing Down in Fast-Paced Game Jams. In *Proceedings of the 8th International Conference on Game Jams, Hackathons and Game Creation Events (ICGJ '24)*. Association for Computing Machinery, New York, NY, USA, 33–40. doi:10.1145/3697789.3697792

[57] Xianzhe Fan, Zihan Wu, Chun Yu, Fenggui Rao, Weinan Shi, and Teng Tu. 2024. ContextCam: Bridging Context Awareness with Creative Human-AI Image Co-Creation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17. doi:10.1145/3613904.3642129

[58] Daisy Fancourt and Saoirse Finn. 2019. What is the evidence on the role of the arts in improving health and well-being? A scoping review. *Health Evidence Network synthesis report* 67 (November 2019). https://www.who.int/europe/publications/i/item/9789289054553

[59] Corey Ford and Nick Bryan-Kinns. 2023. Towards a Reflection in Creative Experience Questionnaire. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16. doi:10.1145/3544548.3581077

[60] Corey Ford, Ashley Noel-Hirst, Sara Cardinale, Jackson Loth, Pedro Sarmento, Elizabeth Wilson, Lewis Wolstanholme, Kyle Worrall, and Nick Bryan-Kinns. 2024. Reflection Across AI-based Music Composition. In *Proceedings of the 16th Conference on Creativity & Cognition*. 398–412. doi:10.1145/3635636.3656185

[61] Jonas Frich, Lindsay MacDonald Vermeulen, Christian Remy, Michael Mose Biskjaer, and Peter Dalsgaard. 2019. Mapping the Landscape of Creativity Support Tools in HCI. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–18. doi:10.1145/3290605.3300619

[62] Jonas Frich, Michael Mose Biskjaer, and Peter Dalsgaard. 2018. Twenty Years of Creativity Research in Human-Computer Interaction: Current State and Future Directions. In *Proceedings of the 2018 Designing Interactive Systems Conference*. 1235–1257. doi:10.1145/3196709.3196732

[63] Jonas Frich, Midas Nouwens, Kim Halskov, and Peter Dalsgaard. 2021. How Digital Tools Impact Convergent and Divergent Thinking in Design Ideation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–11. doi:10.1145/3411764.3445062

[64] Kaori Fujinami, Mami Kosaka, and Bipin Indurkhya. 2018. Painting an Apple with an Apple: A Tangible Tabletop Interface for Painting with Physical Objects. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 4 (2018), 1–22. doi:10.1145/3287040

[65] Ka Yan Fung, Lik Hang Lee, Huamin Qu, Yuelu Li, Shenghui Song, and David Yip. 2024. Create-to-learn Paradigm: A Proxy Visual Storytelling Tool (PVST) for Stimulating Children's Story Sense and Structure. In *Proceedings of the 17th International Symposium on Visual Information Communication and Interaction*. 1–8. doi:10.1145/3678698.3678699

[66] Aleksandra Gajda, Maciej Karwowski, and Ronald A Beghetto. 2017. Creativity and Academic Achievement: A Meta-Analysis. *Journal of Educational Psychology* 109, 2 (2017), 269. doi:10.1037/edu0000133

[67] Katy Ilonka Gero, Vivian Liu, and Lydia Chilton. 2022. Sparks: Inspiration for Science Writing using Language Models. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference*. 1002–1019. doi:10.1145/3532106.3533533

[68] Maliheh Ghajargar, Jeffrey Bardzell, and Love Lagerkvist. 2022. A Redhead Walks into a Bar: Experiences of Writing Fiction with Artificial Intelligence. In *Proceedings of the 25th International Academic MindTrek Conference*. 230–241. doi:10.1145/3569219.3569418

[69] Victor Girotto, Erin Walker, and Winslow Burleson. 2017. The Effect of Peripheral Micro-tasks on Crowd Ideation. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 1843–1854. doi:10.1145/3025453.3025464

[70] Victor Girotto, Erin Walker, and Winslow Burleson. 2019. CrowdMuse: Supporting Crowd Idea Generation through User Modeling and Adaptation. In *Proceedings of the 2019 Conference on Creativity and Cognition*. 95–106. doi:10.1145/3325480.3325497

[71] Andreas Göldi, Thiemo Wambsganss, Seyed Parsa Neshaei, and Roman Rietsche. 2024. Intelligent Support Engages Writers Through Relevant Cognitive Processes. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3613904.3642549

[72] Frederica Gonçalves and Pedro Campos. 2017. Understanding and Evaluating the User Interface Design for Creative Writing. In *Proceedings of the European Conference on Cognitive Ergonomics*. 85–92. doi:10.1145/3121283.3121298

[73] Frederica Gonçalves, Pedro Campos, Julian Hanna, and Simone Ashby. 2015. You're the Voice: Evaluating User Interfaces for Encouraging Underserved Youths to express themselves through Creative Writing. In *Proceedings of the 2015 ACM SIGCHI Conference on Creativity and Cognition*. 63–72. doi:10.1145/2757226.2757236

[74] Frederica Gonçalves, Ana Caraban, Evangelos Karapanos, and Pedro Campos. 2017. What Shall I Write Next?: Subliminal and Supraliminal Priming as Triggers for Creative Writing. In *Proceedings of the European Conference on Cognitive Ergonomics*. 77–84. doi:10.1145/3121283.3121294

[75] Herbert Paul Grice. 1975. Logic and Conversation. *Syntax and Semantics* 3 (1975), 43–58.

[76] Shihui Guo, Yubin Shi, Pintong Xiao, Yinan Fu, Juncong Lin, Wei Zeng, and Tong-Yee Lee. 2023. Creative and Progressive Interior Color Design with Eye-tracked User Preference. *ACM transactions on computer-human interaction* 30, 1 (2023), 1–31. doi:10.1145/3542922

[77] Marte Hoff Hagen, Daniela Soares Cruzes, Letizia Jaccheri, and Jerry Alan Fails. 2023. Evaluating digital creativity support for children: A systematic literature review. *International Journal of Child-Computer Interaction* (2023), 100603. doi:10.1016/j.ijcci.2023.100603

[78] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. *Human Mental Workload* 52 (1988), 139–183. doi:10.1016/S0166-4115(08)62386-9

[79] Rui He, Huaxin Wei, and Ying Cao. 2024. An Interactive System for Supporting Creative Exploration of Cinematic Composition Designs. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–15. doi:10.1145/3654777.3676393

[80] Lena Hegemann, Niraj Ramesh Dayama, Abhishek Iyer, Erfan Farhadi, Ekaterina Marchenko, and Antti Oulasvirta. 2023. CoColor: Interactive Exploration of Color Designs. In *Proceedings of the 28th International Conference on Intelligent User Interfaces*. 106–127. doi:10.1145/3581641.3584089

[81] Jennifer L Heyman, Steven R Rick, Gianni Giacomelli, Haoran Wen, Robert Laubacher, Nancy Taubenslag, Max Knicker, Younes Jeddi, Pranav Ragupathy, Jared Curhan, et al. 2024. Supermind Ideator: How Scaffolding Human-AI Collaboration Can Increase Creativity. In *Proceedings of the ACM Collective Intelligence Conference*. 18–28. doi:10.1145/3643562.3672611

[82] Philipp Pascal Hoffmann, Rina R Wehbe, Max Mühlhäuser, and Mayra Donaji Barrera Machuca. 2024. ThermalPen: Investigating the Influence of Thermal Haptic Feedback for Creativity in 3D Sketching. In *Proceedings of the 16th Conference on Creativity & Cognition*. 661–673. doi:10.1145/3635636.3656191

[83] Md Naimul Hoque, Bhavya Ghai, Kari Kraus, and Niklas Elmqvist. 2023. Portrayal: Leveraging NLP and Visualization for Analyzing Fictional Characters. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 74–94. doi:10.1145/3563657.3596000

[84] Yaxin Hu, Lingjie Feng, Bilge Mutlu, and Henny Admoni. 2021. Exploring the Role of Social Robot Behaviors in a Creative Activity. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference*. 1380–1389. doi:10.1145/3461778.3462116

[85] Mina Huh, Yi-Hao Peng, and Amy Pavel. 2023. GenAssist: Making Image Generation Accessible. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–17. doi:10.1145/3586183.3606735

[86] Angel Hsing-Chi Hwang and Andrea Stevenson Won. 2021. IdeaBot: Investigating Social Facilitation in Human-Machine Team Creativity. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–16. doi:10.1145/3411764.3445270

[87] Rune Møberg Jacobsen, Samuel Rhys Cox, Carla F Griggio, and Niels van Berkel. 2025. Chatbots for Data Collection in Surveys: A Comparison of Four Theory-Based Interview Probes. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. doi:10.1145/3706598.3714128

[88] Anniek Jansen and Sara Colombo. 2023. Mix & Match Machine Learning: An Ideation Toolkit to Design Machine Learning-Enabled Solutions. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–18. doi:10.1145/3569009.3572739

[89] Hans G Jellen and Klaus K Urban. 1986. The TCT-DP (Test for Creative Thinking-Drawing Production): An instrument that can be applied to most age and ability groups. *Creative Child & Adult Quarterly* (1986).

[90] Youngseung Jeon, Seungwan Jin, Patrick C Shih, and Kyungsik Han. 2021. FashionQ: An AI-Driven Creativity Support Tool for Facilitating Ideation in Fashion Design. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–18. doi:10.1145/3411764.3445093

[91] Yunwoo Jeong, Hyungjun Cho, Taewan Kim, and Tek-Jin Nam. 2023. AutomataStage: an AR-mediated Creativity Support Tool for Hands-on Multidisciplinary Learning. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16. doi:10.1145/3544548.3581408

[92] Kevin John, Yinan Li, and Hasti Seifi. 2024. AdapTics: A Toolkit for Creative Design and Integration of Real-Time Adaptive Mid-Air Ultrasound Tactons. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–15. doi:10.1145/3613904.3642090

[93] Girija Kaimal and Kendra Ray and. 2017. Free art-making in an art therapy open studio: changes in affect and self-efficacy. *Arts & Health* 9, 2 (2017), 154–166. doi:10.1080/17533015.2016.1217248

[94] Masahiro Kamihira, Juro Hosoi, Yuki Ban, and Shin'ichi Warisawa. 2024. Multiplexed VR: Individualized Multiplexing Virtual Environment to Facilitate Switches for Group Ideation Creativity. In *Proceedings of the Augmented Humans International Conference 2024*. 168–180. doi:10.1145/3652920.3652935

[95] Youwen Kang, Zhida Sun, Sitong Wang, Zeyu Huang, Ziming Wu, and Xiaojuan Ma. 2021. MetaMap: Supporting Visual Metaphor Ideation through Multi-dimensional Example-based Exploration. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–15. doi:10.1145/3411764.3445325

[96] Hansika Kapoor, Tim Patston, David H. Cropley, Sarah Rezaei, and James C. Kaufman. 2025. Creativity predicts standardized educational outcomes beyond GPA and personality. *Thinking Skills and Creativity* 56 (2025), 101751. doi:10.1016/j.tsc.2024.101751

[97] Hasindu Kariyawasam, Amashi Niwarthana, Alister Palmer, Judy Kay, and Anusha Withana. 2024. Appropriate Incongruity Driven Human-AI Collaborative Tool to Assist Novices in Humorous Content Generation. In *Proceedings of the 29th International Conference on Intelligent User Interfaces*. 650–659. doi:10.1145/3640543.3645161

[98] Jun Kato and Masataka Goto. 2017. f3.js: A Parametric Design Tool for Physical Computing Devices for Both Interaction Designers and End-users. In *Proceedings of the 2017 Conference on Designing Interactive Systems*. 1099–1110. doi:10.1145/3064663.3064681

[99] Jun Kato, Tomoyasu Nakano, and Masataka Goto. 2015. TextAlive: Integrated Design Environment for Kinetic Typography. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 3403–3412. doi:10.1145/2702123.2702140

[100] Joy Kim, Mira Dontcheva, Wilmot Li, Michael S Bernstein, and Daniela Steinsapir. 2015. Motif: Supporting Novice Creativity through Expert Patterns. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 1211–1220. doi:10.1145/2702123.2702507

[101] Jeongyeon Kim, Sangho Suh, Lydia B Chilton, and Haijun Xia. 2023. Metaphorian: Leveraging Large Language Models to Support Extended Metaphor Creation for Science Writing. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 115–135. doi:10.1145/3563657.3595996

[102] Kevin Gonyop Kim, Richard Lee Davis, Alessia Eletta Coppi, Alberto Cattaneo, and Pierre Dillenbourg. 2022. Mixplorer: Scaffolding Design Space Exploration through Genetic Recombination of Multiple Peoples' Designs to Support Novices' Creativity. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–13. doi:10.1145/3491102.3501854

[103] Maria Kjærup, Mikael B. Skov, Peter Axel Nielsen, Jesper Kjeldskov, Jens Gerken, and Harald Reiterer. 2021. *Longitudinal Studies in HCI Research: A Review of CHI Publications From 1982–2019*. Springer International Publishing, Cham, 11–39. doi:10.1007/978-3-030-67322-2_2

[104] Predrag Klasnja, Sunny Consolvo, and Wanda Pratt. 2011. How to evaluate technologies for health behavior change in HCI research. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) *(CHI '11)*. Association for Computing Machinery, New York, NY, USA, 3063–3072. doi:10.1145/1978942.1979396

[105] Janin Koch, Andrés Lucero, Lena Hegemann, and Antti Oulasvirta. 2019. May AI?: Design Ideation with Cooperative Contextual Bandits. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3290605.3300863

[106] Janin Koch, Nicolas Taffin, Michel Beaudouin-Lafon, Markku Laine, Andrés Lucero, and Wendy E Mackay. 2020. ImageSense: An Intelligent Collaborative Ideation Tool to Support Diverse Human-Computer Partnerships. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW1 (2020), 1–27. doi:10.1145/3392850

[107] Janin Koch, Nicolas Taffin, Andrés Lucero, and Wendy E Mackay. 2020. SemanticCollage: Enriching Digital Mood Board Design with Semantic Labels. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference*. 407–418. doi:10.1145/3357236.3395494

[108] Naja Kathrine Kollerup, Maria-Theresa Bahodi, Samuel Rhys Cox, and Niels van Berkel. 2025. Enhancing Self-Efficacy in Health Self-Examination through Conversational Agent's Encouragement. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. doi:10.1145/3706598.3713142

[109] Bettina Laugwitz, Theo Held, and Martin Schrepp. 2008. Construction and Evaluation of a User Experience Questionnaire. In *HCI and Usability for Education and Work: 4th Symposium of the Workgroup Human-Computer Interaction and Usability Engineering of the Austrian Computer Society, USAB 2008, Graz, Austria, November 20-21, 2008. Proceedings 4*. Springer, 63–76.

[110] Tomas Lawton, Francisco J Ibarrola, Dan Ventura, and Kazjon Grace. 2023. Drawing with Reframer: Emergence and Control in Co-Creative AI. In *Proceedings of the 28th International Conference on Intelligent User Interfaces*. 264–277. doi:10.1145/3581641.3584095

[111] Jingyi Li, Eric Rawn, Jacob Ritchie, Jasper Tran O'Leary, and Sean Follmer. 2023. Beyond the Artifact: Power as a Lens for Creativity Support Tools. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–15. doi:10.1145/3586183.3606831

[112] Zhuoyang Li, Minhui Liang, Ray Lc, and Yuhan Luo. 2024. StayFocused: Examining the Effects of Reflective Prompts and Chatbot Support on Compulsive Smartphone Use. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–19. doi:10.1145/3613904.3642479

[113] Antonios Liapis, Christian Guckelsberger, Jichen Zhu, Casper Harteveld, Simone Kriglstein, Alena Denisova, Jeremy Gow, and Mike Preuss. 2023. Designing for Playfulness in Human-AI Authoring Tools. In *Proceedings of the 18th International Conference on the Foundations of Digital Games*. 1–4. doi:10.1145/3582437.3587192

[114] Yuyu Lin, Jiahao Guo, Yang Chen, Cheng Yao, and Fangtian Ying. 2020. It Is Your Turn: Collaborative Ideation With a Co-Creative Robot through Sketch. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14. doi:10.1145/3313831.3376258

[115] Long Ling, Xinyi Chen, Ruoyu Wen, Toby Jia-Jun Li, and RAY LC. 2024. Sketchar: Supporting Character Design and Illustration Prototyping Using Generative AI. *Proceedings of the ACM on Human-Computer Interaction* 8, CHI PLAY (2024), 337. doi:10.1145/3677102

[116] Vivian Liu, Han Qiao, and Lydia Chilton. 2022. Opal: Multimodal Image Generation for News Illustration. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–17. doi:10.1145/3526113.3545621

[117] Vivian Liu, Jo Vermeulen, George Fitzmaurice, and Justin Matejka. 2023. 3DALL-E: Integrating Text-to-Image AI in 3D Design Workflows. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 1955–1977. doi:10.1145/3563657.3596098

[118] James Derek Lomas, Mihovil Karac, and Mathieu Gielen. 2021. Design Space Cards: Using a Card Deck to Navigate the Design Space of Interactive Play. *Proceedings of the ACM on Human-Computer Interaction* 5, CHI PLAY (2021), 1–21. doi:10.1145/3474654

[119] Zhicong Lu, Rubaiat Habib Kazi, Li-yi Wei, Mira Dontcheva, and Karrie Karahalios. 2021. StreamSketch: Exploring Multi-Modal Interactions in Creative Live Streams. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–26. doi:10.1145/3449132

[120] Arnold M Lund. 2001. Measuring Usability with the USE Questionnaire. *Usability Interface* 8, 2 (2001), 3–6.

[121] Caroline Lundqvist, Daniel Klinkhammer, Kim Halskov, Stefan Paul Feyer, Jeanette Falk Olesen, Nanna Inie, Harald Reiterer, and Peter Dalsgaard. 2018. Physical, digital, and hybrid setups supporting card-based collaborative design ideation. In *Proceedings of the 10th Nordic Conference on Human-Computer Interaction*. 260–272. doi:10.1145/3240167.3240177

[122] Eryn Ma, Priya Dixit, Andy Han, Nicholas Marsano, Brooke Sparks, Ashley Sun, Lucas Tiangco, Tongyu Zhou, Jeff Huang, and Alexandra Papoutsaki. 2025. EyeDraw: Investigating the Perceived Effects of Shared Gaze on Remote Collaborative Drawing. *Proc. ACM Hum.-Comput. Interact.* 9, 1, Article GROUP17 (Jan. 2025), 24 pages. doi:10.1145/3701196

[123] Stephen MacNeil, Zijian Ding, Kexin Quan, Thomas j Parashos, Yajie Sun, and Steven P Dow. 2021. Framing Creative Work: Helping Novices Frame Better Problems through Interactive Scaffolding. In *Proceedings of the 13th Conference on Creativity and Cognition*. 1–10. doi:10.1145/3450741.3465261

[124] Neil Maiden, Konstantinos Zachos, Amanda Brown, George Brock, Lars Nyre, Aleksander Nygård Tonheim, Dimitris Apsotolou, and Jeremy Evans. 2018. Making the News: Digital Creativity Support for Journalists. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) *(CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3173574.3174049

[125] Neil Maiden, Konstantinos Zachos, Amanda Brown, Lars Nyre, Balder Holm, Aleksander Nygård Tonheim, Claus Hesseling, Andrea Wagemans, and Dimitris Apostolou. 2019. Evaluating the Use of Digital Creativity Support by Journalists in Newsrooms. In *Proceedings of the 2019 Conference on Creativity and Cognition*. 222–232. doi:10.1145/3325480.3325484

[126] Neil Maiden, Konstantinos Zachos, James Lockerbie, Sergio Levis, Kasia Camargo, Shaun Hoddy, and Gianluca Allemandi. 2017. Evaluating Digital Creativity Support To Improve Health-and-Safety in a Manufacturing Plant. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 7005–7014. doi:10.1145/3025453.3025712

[127] Daniel Manesh, Douglas Bowman Jr, and Sang Won Lee. 2024. SHARP: Exploring Version Control Systems in Live Coding Music. In *Proceedings of the 16th Conference on Creativity & Cognition*. 426–437. doi:10.1145/3635636.3656195

[128] Manon Marinussen and Alwin de Rooij. 2019. Being Yourself to be Creative: How Self-Similar Avatars can Support the Generation of Original Ideas in Virtual Environments. In *Proceedings of the 2019 Conference on Creativity and Cognition*. 285–293. doi:10.1145/3325480.3325482

[129] Charles Martin, Henry Gardner, Ben Swift, and Michael Martin. 2016. Intelligent Agents and Networked Buttons Improve Free-Improvised Ensemble Music-Making on Touch-Screens. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 2295–2306. doi:10.1145/2858036.2858269

[130] Janetta Mitchell McCoy and Gary W Evans. 2002. The Potential Role of the Physical Environment in Fostering Creativity. *Creativity Research Journal* 14, 3-4 (2002), 409–426. doi:10.1207/S15326934CRJ1434_11

[131] Elisa D Mekler and Kasper Hornbæk. 2019. A Framework for the Experience of Meaning in Human-Computer Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–15. doi:10.1145/3290605.3300455

[132] Liang Men and Nick Bryan-Kinns. 2019. LeMo: Exploring Virtual Space for Collaborative Creativity. In *Proceedings of the 2019 Conference on Creativity and Cognition*. 71–82. doi:10.1145/3325480.3325495

[133] Piotr Mirowski, Kory W Mathewson, Jaylen Pittman, and Richard Evans. 2023. Co-Writing Screenplays and Theatre Scripts with Language Models: Evaluation by Industry Professionals. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–34. doi:10.1145/3544548.3581225

[134] Timothy Neate, Abi Roper, Stephanie Wilson, and Jane Marshall. 2019. Empowering Expression for Users with Aphasia through Constrained Creativity. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3290605.3300615

[135] Timothy Neate, Abi Roper, Stephanie Wilson, Jane Marshall, and Madeline Cruice. 2020. CreaTable Content and Tangible Interaction in Aphasia. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14. doi:10.1145/3313831.3376490

[136] Tricia J Ngoon, Joy O Kim, and Scott Klemmer. 2021. Shöwn: Adaptive Conceptual Guidance Aids Example Use in Creative Tasks. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference*. 1834–1845. doi:10.1145/3461778.3462072

[137] Zheng Ning, Zheng Zhang, Jerrick Ban, Kaiwen Jiang, Ruohong Gan, Yapeng Tian, and Toby Jia-Jun Li. 2024. MIMOSA: Human-AI Co-Creation of Computational Spatial Audio Effects on Videos. In *Proceedings of the 16th Conference on Creativity & Cognition*. 156–169. doi:10.1145/3635636.3656189

[138] Sascha Oberhuber, Tina Kothe, Stefan Schneegass, and Florian Alt. 2017. Augmented Games: Exploring Design Opportunities in AR Settings With Children. In *Proceedings of the 2017 Conference on Interaction Design and Children*. 371–377. doi:10.1145/3078072.3079734

[139] Changhoon Oh, Jungwoo Song, Jinhan Choi, Seonghyeon Kim, Sungwoo Lee, and Bongwon Suh. 2018. I Lead, You Help but Only with Enough Details: Understanding User Experience of Co-Creation with Artificial Intelligence. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–13. doi:10.1145/3173574.3174223

[140] Jeongseok Oh, Seungju Kim, and Seungjun Kim. 2024. LumiMood: A Creativity Support Tool for Designing the Mood of a 3D Scene. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–21. doi:10.1145/3613904.3642440

[141] Jonas Oppenlaender, Thanassis Tiropanis, and Simo Hosio. 2020. CrowdUI: Supporting Web Design with the Crowd. *Proceedings of the ACM on Human-Computer Interaction* 4, EICS (2020), 1–28. doi:10.1145/3394978

[142] Srishti Palani, David Ledo, George Fitzmaurice, and Fraser Anderson. 2022. "I don't want to feel like I'm working in a 1960s factory": The Practitioner Perspective on Creativity Support Tool Adoption. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–18. doi:10.1145/3491102.3501933

[143] Srishti Palani and Gonzalo Ramos. 2024. Evolving Roles and Workflows of Creative Practitioners in the Age of Generative AI. In *Proceedings of the 16th Conference on Creativity & Cognition*. 170–184. doi:10.1145/3635636.3656190

[144] Rock Yuren Pang, Hope Schroeder, Kynnedy Simone Smith, Solon Barocas, Ziang Xiao, Emily Tseng, and Danielle Bragg. 2025. Understanding the LLM-ification of CHI: Unpacking the Impact of LLMs at CHI through a Systematic Literature Review. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. doi:10.48550/arXiv.2501.12557

[145] Jung Wook Park, Sienna Xin Sun, Tingyu Cheng, Dong Whi Yoo, Jiawei Zhou, Youngwook Do, Gregory D Abowd, and Rosa I Arriaga. 2023. Exergy: A Toolkit to Simplify Creative Applications of Wind Energy Harvesting. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 1 (2023), 1–28. doi:10.1145/3580814

[146] Xiaohan Peng, Janin Koch, and Wendy E Mackay. 2024. DesignPrompt: Using Multimodal Interaction for Design Exploration with Generative AI. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 804–818. doi:10.1145/3643834.3661588

[147] Savvas Petridis, Nicholas Diakopoulos, Kevin Crowston, Mark Hansen, Keren Henderson, Stan Jastrzebski, Jeffrey V Nickerson, and Lydia B Chilton. 2023. AngleKindling: Supporting Journalistic Angle Ideation with Large Language Models. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16. doi:10.1145/3544548.3580907

[148] Cecil Piya, Vinayak, Senthil Chandrasegaran, Niklas Elmqvist, and Karthik Ramani. 2017. Co-3Deator: A Team-First Collaborative 3D Design Ideation Tool. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 6581–6592. doi:10.1145/3025453.3025825

[149] Jonathan A Plucker, Ronald A Beghetto, and Gayle T Dow. 2004. Why Isn't Creativity More Important to Educational Psychologists? Potentials, Pitfalls, and Future Directions in Creativity Research. *Educational Psychologist* 39, 2 (2004), 83–96. doi:10.1207/s15326985ep3902_1

[150] Leonard Przybilla, Luka Baar, Manuel Wiesche, and Helmut Krcmar. 2019. Machines as Teammates in Creative Teams: Digital Facilitation of the Dual Pathways to Creativity. In *Proceedings of the 2019 on Computers and People Research Conference*. 94–102. doi:10.1145/3322385.3322402

[151] Bijisa Pyakurel, Shreyosi Endow, Carlos Donjuan, and Cesar Torres. 2023. Cogcues: Shifting Perception through Interactive Projected Cues in Still Life Drawing. In *Proceedings of the 15th Conference on Creativity and Cognition*. 171–184. doi:10.1145/3591196.3593332

[152] Hua Xuan Qin, Shan Jin, Ze Gao, Mingming Fan, and Pan Hui. 2024. CharacterMeet: Supporting Creative Writers' Entire Story Character Construction Processes Through Conversation with LLM-Powered Chatbot Avatars. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–19. doi:10.1145/3613904.3642105

[153] Amon Rapp, Lorenzo Curti, and Arianna Boldi. 2021. The human side of human-chatbot interaction: A systematic literature review of ten years of research on text-based chatbots. *International Journal of Human-Computer Studies* 151 (2021), 102630. doi:10.1016/j.ijhcs.2021.102630

[154] Christian Remy, Lindsay MacDonald Vermeulen, Jonas Frich, Michael Mose Biskjaer, and Peter Dalsgaard. 2020. Evaluating Creativity Support Tools in HCI Research. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference*. 457–476. doi:10.1145/3357236.3395474

[155] Mohi Reza, Nathan M Laundry, Ilya Musabirov, Peter Dushniku, Zhi Yuan "Michael" Yu, Kashish Mittal, Tovi Grossman, Michael Liut, Anastasia Kuzminykh, and Joseph Jay Williams. 2024. ABScribe: Rapid Exploration & Organization of Multiple Writing Variations in Human-AI Co-Writing Tasks using Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18. doi:10.1145/3613904.3641899

[156] Jeba Rezwana and Mary Lou Maher. 2022. Understanding User Perceptions, Collaborative Experience and User Engagement in Different Human-AI Interaction Designs for Co-Creative Systems. In *Proceedings of the 14th Conference on Creativity and Cognition*. 38–48. doi:10.1145/3527927.3532789

[157] Jeba Rezwana and Mary Lou Maher. 2023. Designing Creative AI Partners with COFI: A Framework for Modeling Interaction in Human-AI Co-Creative Systems. *ACM Transactions on Computer-Human Interaction* 30, 5 (2023), 1–28. doi:10.1145/3519026

[158] Ruben Schlagowski, Silvan Mertes, Dariia Nazarenko, Alexander Dauber, and Elisabeth André. 2024. XR Composition in the Wild: The Impact of User Environments on Creativity, UX and Flow during Music Production in Augmented Reality. In *Proceedings of the 19th International Audio Mostly Conference: Explorations in Sonic Cultures*. 152–161. doi:10.1145/3678299.3678314

[159] Ruben Schlagowski, Fabian Wildgrube, Silvan Mertes, Ceenu George, and Elisabeth André. 2022. Flow with the Beat! Human-Centered Design of Virtual Environments for Musical Creativity Support in VR. In *Proceedings of the 14th Conference on Creativity and Cognition*. 428–442. doi:10.1145/3527927.3532799

[160] Oliver Schmitt and Daniel Buschek. 2021. CharacterChat: Supporting the Creation of Fictional Characters through Conversation and Progressive Manifestation with a Chatbot. In *Proceedings of the 13th Conference on Creativity and Cognition*. 1–10. doi:10.1145/3450741.3465253

[161] Ralf Schwarzer and Matthias Jerusalem. 1995. Generalized Self-Efficacy Scale. *J. Weinman, S. Wright, & M. Johnston, Measures in Health Psychology: A User's Portfolio. Causal and Control Beliefs* 35, 37 (1995), 82–003.

[162] Jasjeet Singh Seehra, Ansh Verma, Kylie Peppler, and Karthik Ramani. 2015. HandiMate: Create and Animate using Everyday Objects as Material. In *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction*. 117–124. doi:10.1145/2677199.2680570

[163] Yulin Shen, Yifei Shen, Jiawen Cheng, Chutian Jiang, Mingming Fan, and Zeyu Wang. 2024. Neural Canvas: Supporting Scenic Design Prototyping by Integrating 3D Sketching and Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18. doi:10.1145/3613904.3642096

[164] Yang Shi, Nan Cao, Xiaojuan Ma, Siji Chen, and Pei Liu. 2020. EmoG: Supporting the Sketching of Emotional Expressions for Storyboarding. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3313831.3376520

[165] Evan Shimizu, Matthew Fisher, Sylvain Paris, James McCann, and Kayvon Fatahalian. 2020. Design Adjectives: A Framework for Interactive Model-Guided Exploration of Parameterized Design Spaces. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*. 261–278. doi:10.1145/3379337.3415866

[166] Maria Shugrina, Jingwan Lu, and Stephen Diverdi. 2017. Playful palette: an interactive parametric color mixer for artists. *ACM Transactions on Graphics (TOG)* 36, 4 (2017), 1–10. doi:10.1145/3072959.3073690

[167] Maria Shugrina, Wenjia Zhang, Fanny Chevalier, Sanja Fidler, and Karan Singh. 2019. Color Builder: A Direct Manipulation Interface for Versatile Color Theme Authoring. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3290605.3300686

[168] Pao Siangliulue, Kenneth C Arnold, Krzysztof Z Gajos, and Steven P Dow. 2015. Toward Collaborative Ideation at Scale: Leveraging Ideas from Others to Generate More Creative and Diverse Ideas. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing*. 937–945. doi:10.1145/2675133.2675239

[169] Pao Siangliulue, Joel Chan, Steven P Dow, and Krzysztof Z Gajos. 2016. IdeaHound: Improving Large-scale Collaborative Ideation with Crowd-Powered Real-time Semantic Modeling. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology*. 609–624. doi:10.1145/2984511.2984578

[170] Pao Siangliulue, Joel Chan, Krzysztof Z Gajos, and Steven P Dow. 2015. Providing Timely Examples Improves the Quantity and Quality of Generated Ideas. In *Proceedings of the 2015 ACM SIGCHI Conference on Creativity and Cognition*. 83–92. doi:10.1145/2757226.2757230

[171] Nikhil Singh, Guillermo Bernal, Daria Savchenko, and Elena L Glassman. 2023. Where to Hide a Stolen Elephant: Leaps in Creative Writing with Multimodal Machine Intelligence. *ACM Transactions on Computer-Human Interaction* 30, 5 (2023), 1–57. doi:10.1145/3511599

[172] Kihoon Son, DaEun Choi, Tae Soo Kim, Young-Ho Kim, and Juho Kim. 2024. GenQuery: Supporting Expressive Visual Search with Generative Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–19. doi:10.1145/3613904.3642847

[173] Tianqi Song, Jack Jamieson, Tianwen Zhu, Naomi Yamashita, and Yi-Chieh Lee. 2025. From Interaction to Attitude: Exploring the Impact of Human-AI Cooperation on Mental Illness Stigma. *Proc. ACM Hum.-Comput. Interact.* CSCW1 (2025). doi:10.48550/arXiv.2501.01220

[174] Arvind Srinivasan and Joel Chan. 2024. Improving Selection of Analogical Inspirations through Chunking and Recombination. In *Proceedings of the 16th Conference on Creativity & Cognition*. 374–397. doi:10.1145/3635636.3656207

[175] Evgeny Stemasov, Simon Demharter, Max Rädler, Jan Gugenheimer, and Enrico Rukzio. 2024. pARam: Leveraging Parametric Design in Extended Reality to Support the Personalization of Artifacts for Personal Fabrication. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–22. doi:10.1145/3613904.3642083

[176] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminate: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–26. doi:10.1145/3613904.3642400

[177] Sangho Suh, Sydney Lamorea, Edith Law, and Leah Zhang-Kennedy. 2022. PrivacyToon: Concept-driven Storytelling with Creativity Support for Privacy Concepts. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference*. 41–57. doi:10.1145/3532106.3533557

[178] Nicole Sultanum, Fanny Chevalier, Zoya Bylinskii, and Zhicheng Liu. 2021. Leveraging Text-Chart Links to Support Authoring of Data-Driven Articles with VizFlow. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–17. doi:10.1145/3411764.3445354

[179] Carla Tamburro, Timothy Neate, Abi Roper, and Stephanie Wilson. 2020. Accessible Creativity with a Comic Spin. In *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility*. 1–11. doi:10.1145/3373625.3417012

[180] Carla Tamburro, Timothy Neate, Abi Roper, and Stephanie Wilson. 2022. Comic Spin: A Comic Creation Tool Enabling Self-Expression for People with Aphasia. *ACM Transactions on Accessible Computing (TACCESS)* 15, 2 (2022), 1–27. doi:10.1145/3508500

[181] Robert E Thayer. 1986. Activation–Deactivation Adjective Check List: Current overview and structural analysis. *Psychological reports* 58, 2 (1986), 607–614. doi:10.2466/pr0.1986.58.2.607

[182] Cesar Torres, Wilmot Li, and Eric Paulos. 2016. ProxyPrint: Supporting Crafting Practice through Physical Computational Proxies. In *Proceedings of the 2016 ACM Conference on Designing Interactive Systems*. 158–169. doi:10.1145/2901790.2901828

[183] Georg Volkmar, Dmitry Alexandrovsky, Asmus Eike Eilks, Dirk Queck, Marc Herrlich, and Rainer Malaka. 2022. Effects of PCG on Creativity in Playful City-Building Environments in VR. *Proceedings of the ACM on Human-Computer Interaction* 6, CHI PLAY (2022), 1–20. doi:10.1145/3549493

[184] Samangi Wadinambiarachchi, Ryan M Kelly, Saumya Pareek, Qiushi Zhou, and Eduardo Velloso. 2024. The Effects of Generative AI on Design Fixation and Divergent Thinking. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18. doi:10.1145/3613904.3642919

[185] Nadine Wagener, Leon Reicherts, Nima Zargham, Natalia Bartłomiejczyk, Ava Elizabeth Scott, Katherine Wang, Marit Bentvelzen, Evropi Stefanidi, Thomas Mildner, Yvonne Rogers, et al. 2023. SelVReflect: A Guided VR Experience Fostering Reflection on Personal Challenges. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–17. doi:10.1145/

3544548.3580763
[186] Shaun Wallace, Brendan Le, Luis A Leiva, Aman Haq, Ari Kintisch, Gabrielle Bufrem, Linda Chang, and Jeff Huang. 2020. Sketchy: Drawing Inspiration from the Crowd. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2 (2020), 1–27. doi:10.1145/3415243
[187] Qian Wan, Xin Feng, Yining Bei, Zhiqi Gao, and Zhicong Lu. 2024. Metamorpheus: Interactive, Affective, and Creative Dream Narration Through Metaphorical Visual Storytelling. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–16. doi:10.1145/3613904.3642410
[188] Qian Wan and Zhicong Lu. 2023. GANCollage: A GAN-Driven Digital Mood Board to Facilitate Ideation in Creativity Support. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 136–146. doi:10.1145/3563657.3596072
[189] Sitong Wang, Samia Menon, Tao Long, Keren Henderson, Dingzeyu Li, Kevin Crowston, Mark Hansen, Jeffrey V Nickerson, and Lydia B Chilton. 2024. ReelFramer: Human-AI Co-Creation for News-to-Video Translation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–20. doi:10.1145/3613904.3642868
[190] Sitong Wang, Zheng Ning, Anh Truong, Mira Dontcheva, Dingzeyu Li, and Lydia B Chilton. 2024. PodReels: Human-AI Co-Creation of Video Podcast Teasers. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 958–974. doi:10.1145/3643834.3661591
[191] Sitong Wang, Savvas Petridis, Taeahn Kwon, Xiaojuan Ma, and Lydia B Chilton. 2023. PopBlends: Strategies for Conceptual Blending with Large Language Models. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19. doi:10.1145/3544548.3580948
[192] Tianyi Wang, Ke Huo, Pratik Chawla, Guiming Chen, Siddharth Banerjee, and Karthik Ramani. 2018. Plain2Fun: Augmenting Ordinary Objects with Interactive Functions by Auto-Fabricating Surface Painted Circuits.. In *Conference on Designing Interactive Systems*. 1095–1106. doi:10.1145/3196709.3196791
[193] Zeyu Wang, Cuong Nguyen, Paul Asente, and Julie Dorsey. 2023. PointShopAR: Supporting Environmental Design Prototyping Using Point Cloud in Augmented Reality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–15. doi:10.1145/3544548.3580776
[194] Jeremy Warner, Kyu Won Kim, and Bjoern Hartmann. 2023. Interactive Flexible Style Transfer for Vector Graphics. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–14. doi:10.1145/3586183.3606751
[195] Kento Watanabe, Yuichiroh Matsubayashi, Kentaro Inui, Tomoyasu Nakano, Satoru Fukayama, and Masataka Goto. 2017. LyriSys: An Interactive Support System for Writing Lyrics Based on Topic Transition. In *Proceedings of the 22nd International Conference on Intelligent User Interfaces*. 559–563. doi:10.1145/3025171.3025194
[196] David Watson, Lee Anna Clark, and Auke Tellegen. 1988. Development and Validation of Brief Measures of Positive and Negative Affect: The PANAS Scales. *Journal of personality and social psychology* 54, 6 (1988), 1063. doi:10.1037/0022-3514.54.6.1063
[197] Christoph Johannes Weber, Sebastian Burgkart, and Sylvia Rothe. 2024. wr-AIter: Enhancing Ownership Perception in AI-Driven Script Writing. In *Proceedings of the 2024 ACM International Conference on Interactive Media Experiences*. 145–156. doi:10.1145/3639701.3656325
[198] Joel Wester, Henning Pohl, Simo Hosio, and Niels van Berkel. 2024. "This Chatbot Would Never...": Perceived Moral Agency of Mental Health Chatbots. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1, Article 133 (April 2024), 28 pages. doi:10.1145/3637410
[199] Nora S Willett, Kurt Fleischer, Haldean Brown, Ilene L E, and Mark Meyer. 2023. CurveCrafter: A System for Animated Curve Manipulation. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–11. doi:10.1145/3586183.3606792
[200] Nora S Willett, Wilmot Li, Jovan Popovic, Floraine Berthouzoz, and Adam Finkelstein. 2017. Secondary Motion for Performed 2D Animation. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology*. 97–108. doi:10.1145/3126594.3126641
[201] Randi Williams, Safinah Ali, Raúl Alcantara, Tasneem Burghleh, Sharifa Alghowinem, and Cynthia Breazeal. 2024. Doodlebot: An Educational Robot for Creativity and AI Literacy. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*. 772–780. doi:10.1145/3610977.3634950
[202] Blake Williford, Abhay Doke, Michel Pahud, Ken Hinckley, and Tracy Hammond. 2019. DrawMyPhoto: Assisting Novices in Drawing from Photographs. In *Proceedings of the 2019 Conference on Creativity and Cognition*. 198–209. doi:10.1145/3325480.3325507
[203] Erik Wolf, Sara Klüber, Chris Zimmerer, Jean-Luc Lugrin, and Marc Erich Latoschik. 2019. "Paint that object yellow": Multimodal Interaction to Enhance Creativity During Design Tasks in VR. In *2019 International Conference on Multimodal Interaction*. 195–204. doi:10.1145/3340555.3353724
[204] Di Wu, Mingzhu Wu, Yeye Li, Jianan Jiang, Xinglin Li, Hanhui Deng, Can Liu, and Yi Li. 2024. StyleWe: Towards Style Fusion in Generative Fashion Design with Efficient Federated AI. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW2 (2024), 1–31. doi:10.1145/3687054
[205] Di Wu, Zhiwang Yu, Nan Ma, Jianan Jiang, Yuetian Wang, Guixiang Zhou, Hanhui Deng, and Yi Li. 2023. StyleMe: Towards Intelligent Fashion Generation with Designer Style. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16. doi:10.1145/3544548.3581377
[206] Hui-Yin Wu, Francesca Palù, Roberto Ranon, and Marc Christie. 2018. Thinking Like a Director: Film Editing Patterns for Virtual Cinematographic Storytelling. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)* 14, 4 (2018), 1–22. doi:10.1145/3241057
[207] Yongmeng Wu and Nick Bryan-Kinns. 2017. Supporting Non-Musicians? Creative Engagement with Musical Interfaces. In *Proceedings of the 2017 ACM SIGCHI Conference on Creativity and Cognition*. 275–286. doi:10.1145/3059454.3059457
[208] Shishi Xiao, Liangwei Wang, Xiaojuan Ma, and Wei Zeng. 2024. TypeDance: Creating Semantic Typographic Logos from Image through Personalized Generation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18. doi:10.1145/3613904.3642185
[209] Ziang Xiao, Michelle X Zhou, Wenxi Chen, Huahai Yang, and Changyan Chi. 2020. If I Hear You Correctly: Building and Evaluating Interview Chatbots with Active Listening Skills. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14. doi:10.1145/3313831.3376131
[210] Haoran Xu, Shuyao Chen, and Ying Zhang. 2023. Magical Brush: A Symbol-Based Modern Chinese Painting System for Novices.. In *CHI*. 131–1. doi:10.1145/3544548.3581429
[211] Xiaotong Xu, Srishti Palani, Azzaya Munkhbat, Tiffany Lee, and Steven P Dow. 2024. Idea-Centric Search: Four Patterns of Information Seeking During Creative Ideation. In *Proceedings of the 16th Conference on Creativity & Cognition*. 280–291. doi:10.1145/3635636.3656193
[212] Xiaotong Xu, Rosaleen Xiong, Boyang Wang, David Min, and Steven P Dow. 2021. IdeateRelate: An Examples Gallery That Helps Creators Explore Ideas in Relation to Their Own. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), 1–18. doi:10.1145/3479496
[213] Xiaotong Xu, Jiayu Yin, Catherine Gu, Jenny Mar, Sydney Zhang, Jane L E, and Steven P Dow. 2024. Jamplate: Exploring LLM-Enhanced Templates for Idea Reflection. In *Proceedings of the 29th International Conference on Intelligent User Interfaces*. 907–921. doi:10.1145/3640543.3645196
[214] Chuan Yan, John Joon Young Chung, Yoon Kiheon, Yotam Gingold, Eytan Adar, and Sungsoo Ray Hong. 2022. FlatMagic: Improving Flat Colorization through AI-driven Design for Digital Comic Professionals. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–17. doi:10.1145/3491102.3502075
[215] Zihan Yan, Yanhong Wu, Danli Luo, Chao Zhang, Qihang Jin, Wei Chen, Yingcai Wu, Xiang'Anthony' Chen, Guanyun Wang, and Haipeng Mi. 2023. NaCanva: Exploring and Enabling the Nature-Inspired Creativity for Children. *Proceedings of the ACM on Human-Computer Interaction* 7, MHCI (2023), 1–25. doi:10.1145/3604262
[216] Zihan Yan, Chunxu Yang, Qihao Liang, and Xiang'Anthony' Chen. 2023. XCreation: A Graph-based Crossmodal Generative Creativity Support Tool. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–15. doi:10.1145/3586183.3606826
[217] Zhihao Yao, Yao Lu, Qirui Sun, Shiqing Lyu, Hanxuan Li, Xing-Dong Yang, Xuezhu Wang, Guanhong Liu, and Haipeng Mi. 2024. Lumina: A Software Tool for Fostering Creativity in Designing Chinese Shadow Puppets. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–15. doi:10.1145/3654777.3676426
[218] Yu-Chun Grace Yen, Steven P Dow, Elizabeth Gerber, and Brian P Bailey. 2017. Listen to Others, Listen to Yourself: Combining Feedback Review and Reflection to Improve Iterative Design. In *Proceedings of the 2017 ACM SIGCHI Conference on Creativity and Cognition*. 158–170. doi:10.1145/3059454.3059468
[219] Yu-Chun Grace Yen, Isabelle Yan Pan, Grace Lin, Mingyi Li, Hyoungwook Jin, Mengyi Chen, Haijun Xia, Steven P Dow, et al. 2024. When to Give Feedback: Exploring Tradeoffs in the Timing of Design Feedback. In *Proceedings of the 16th Conference on Creativity & Cognition*. ACM, 292–310. doi:10.1145/3635636.3656183
[220] Niloofar Zarei, Sharon Lynn Chu, Francis Quek, Nanjie 'Jimmy' Rao, and Sarah Anne Brown. 2020. Investigating the Effects of Self-Avatars and Story-Relevant Avatars on Children's Creative Storytelling. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–11. doi:10.1145/3313831.3376331
[221] Chao Zhang, Xuechen Liu, Katherine Ziska, Soobin Jeon, Chi-Lin Yu, and Ying Xu. 2024. Mathemyths: Leveraging Large Language Models to Teach Mathematical Language through Child-AI Co-Creative Storytelling. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–23. doi:10.1145/3613904.3642647
[222] Chao Zhang, Cheng Yao, Jiayi Wu, Weijia Lin, Lijuan Liu, Ge Yan, and Fangtian Ying. 2022. StoryDrawer: A Child–AI Collaborative Drawing System to Support Children's Creative Visual Storytelling. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–15. doi:10.1145/3491102.3501914

[223] Hongbo Zhang, Pei Chen, Xuelong Xie, Zhaoqu Jiang, Zihong Zhou, and Lingyun Sun. 2024. A Hybrid Prototype Method Combining Physical Models and Generative Artificial Intelligence to Support Creativity in Conceptual Design. *ACM Transactions on Computer-Human Interaction* 31, 5 (2024), 1–34. doi:10.1145/3689433

[224] Hongbo Zhang, Pei Chen, Xuelong Xie, Chaoyi Lin, Lianyan Liu, Zhuoshu Li, Weitao You, and Lingyun Sun. 2024. ProtoDreamer: A Mixed-prototype Tool Combining Physical Model and Generative AI to Support Conceptual Design. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–18. doi:10.1145/3654777.3676399

[225] Kejun Zhang, Shixuan Cai, Wenjing Yang, Wenqi Wu, and Hanshu Shen. 2024. Exploring Optimal Combinations: The Impact of Sequential Multimodal Inspirational Stimuli in Design Concepts on Creativity. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 2788–2801. doi:10.1145/3643834.3661501

[226] Zheng Zhang, Jie Gao, Ranjodh Singh Dhaliwal, and Toby Jia-Jun Li. 2023. VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–30. doi:10.1145/3586183.3606800

[227] Qingxiao Zheng, Yiliu Tang, Yiren Liu, Weizi Liu, and Yun Huang. 2022. UX Research on Conversational Human-AI Interaction: A Literature Review of the ACM Digital Library. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–24. doi:10.1145/3491102.3501855

[228] Ruican Zhong, Donghoon Shin, Rosemary Meza, Predrag Klasnja, Lucas Colusso, and Gary Hsieh. 2024. AI-Assisted Causal Pathway Diagram for Human-Centered Design. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–19. doi:10.1145/3613904.3642179

[229] Jiayi Zhou, Renzhong Li, Junxiu Tang, Tan Tang, Haotian Li, Weiwei Cui, and Yingcai Wu. 2024. Understanding Nonlinear Collaboration between Human and AI Agents: A Co-design Framework for Creative Design. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–16. doi:10.1145/3613904.3642812

[230] Zicheng Zhu, Yugin Tan, Naomi Yamashita, Yi-Chieh Lee, and Renwen Zhang. 2025. The Benefits of Prosociality towards AI Agents: Examining the Effects of Helping AI Agents on Human Well-Being. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. doi:10.48550/arXiv.2502.02911

[231] Chen Zhu-Tian, Wai Tong, Qianwen Wang, Benjamin Bach, and Huamin Qu. 2020. Augmenting Static Visualizations with PapARVis Designer. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3313831.3376436

[232] Chris Zimmerer, Erik Wolf, Sara Wolf, Martin Fischbach, Jean-Luc Lugrin, and Marc Erich Latoschik. 2020. Finally on Par?! Multimodal and Unimodal Interaction for Open Creative Design Tasks in Virtual Reality. In *Proceedings of the 2020 International Conference on Multimodal Interaction*. 222–231. doi:10.1145/3382507.3418850